

Lab 00

How to Survive & Introduction to Git

Software Studio
DataLab, CS, NTHU

Teaching Assistants



Chi-Yu Li

李其祐



Yu-Chun Hsiao

蕭仔君



Shih-Ping Huang

黃詩評

How to Find Us?

- Office Hour (TAs)
 - Thu. 10:10am-12:00pm at Delta 724
- Email
 - ssta2026@datalab.cs.nthu.edu.tw
- Online Forum
 - eeclass

If I have Question?



If I have Question?

- Always **Google or ask Chatgpt first!**
 - Learn how to google and generative AI is important.
- If you try your best but still can't catch it.
- Feel free to ask us on eeclash or office hour



Today's exercise

- Install Git command line tool in your computer
 - Follow appendix “Git Command-line Tool Installation”.
- Try to submit in GitLab.
- Set up flutter environment (If we have time)

Outline

- General Rule
- Introduction to Git
- Version control
- Branch and merge
- How to Submit Your Code to Gitlab
- Tools & References

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The Policy of Labs

- All labs need to be submitted to GitLab.
- Late submission will **not** be accepted.
- Plagiarism will not be tolerated.
 - If we find you copy someone's code, you will get **0 point** for that lab.
- Grading
 - Submission before lab ends gets 100% score
 - Submission before **11:59pm** gets 60% score

Grading Example

- 4 problems, 25% each
- Solved 4 during the lab
 - 100
- Solved 3 during the lab, 1 before 11:59pm
 - $75 + 25 * 0.6 = 90$
- Solved 4 after the lab, before 11:59pm
 - $100 * 0.6 = 60$

Outline

- General Rule
- Introduction to Git
- Version control
- Branch and merge
- How to Submit Your Code to Gitlab
- Tools & References

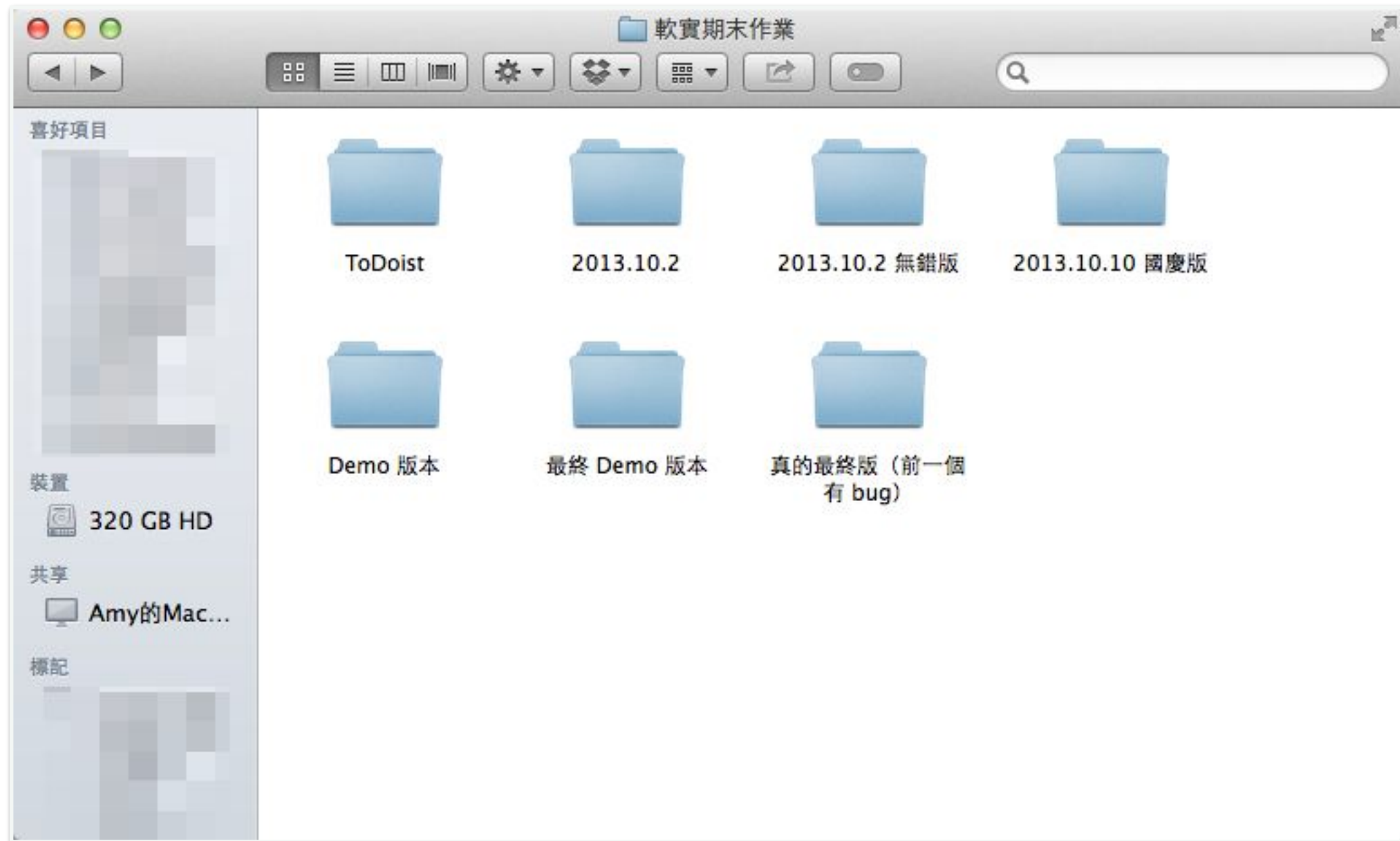
Why should we use
version control?

We want to track what we
did and when we did it.

Why use version control system?

- Managing your projects - tracking your files and modifications.
- Synchronization between modifications made by different developers.
- Revision history is still very helpful even if you work alone.

Students' VCS



Git



Git

- Git is a popular version control system which is
 - Fast
 - Easy to use
 - Distributed
- A git repository is a mini database that tracks your files.

Be Professional



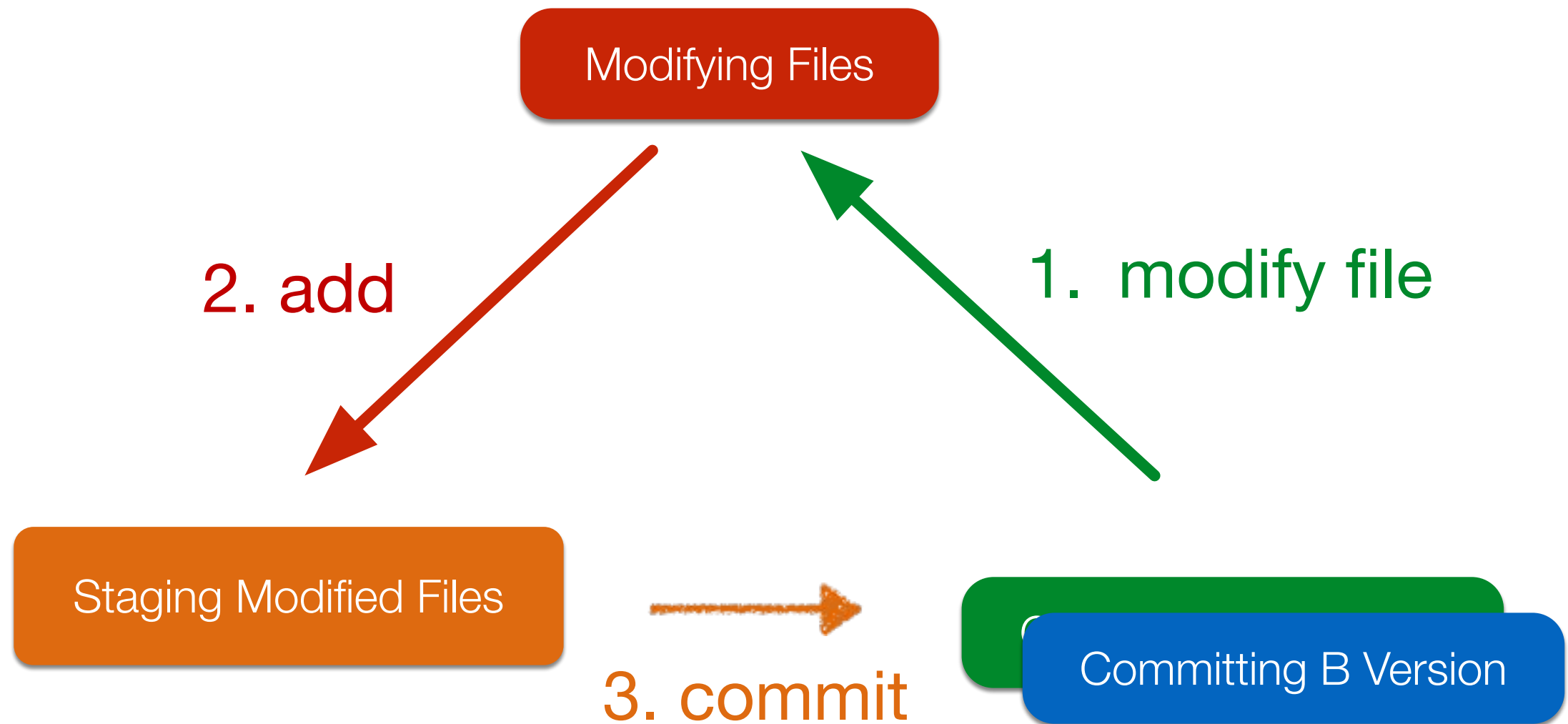
Outline

- General Rule
- Introduction to Git
- **Version control**
- Branch and merge
- How to Submit Your Code to Gitlab
- Tools & References

Git VCS (1/2)

- With a local repository in your computer, you'll need following operations to make git track your work:
 1. Create/modify files
 2. Let git monitor the files by *adding* them to staging files.
 3. *Commit* your changes to and git will create snapshots (versions) of the files for you.

Git VCS (2/2)

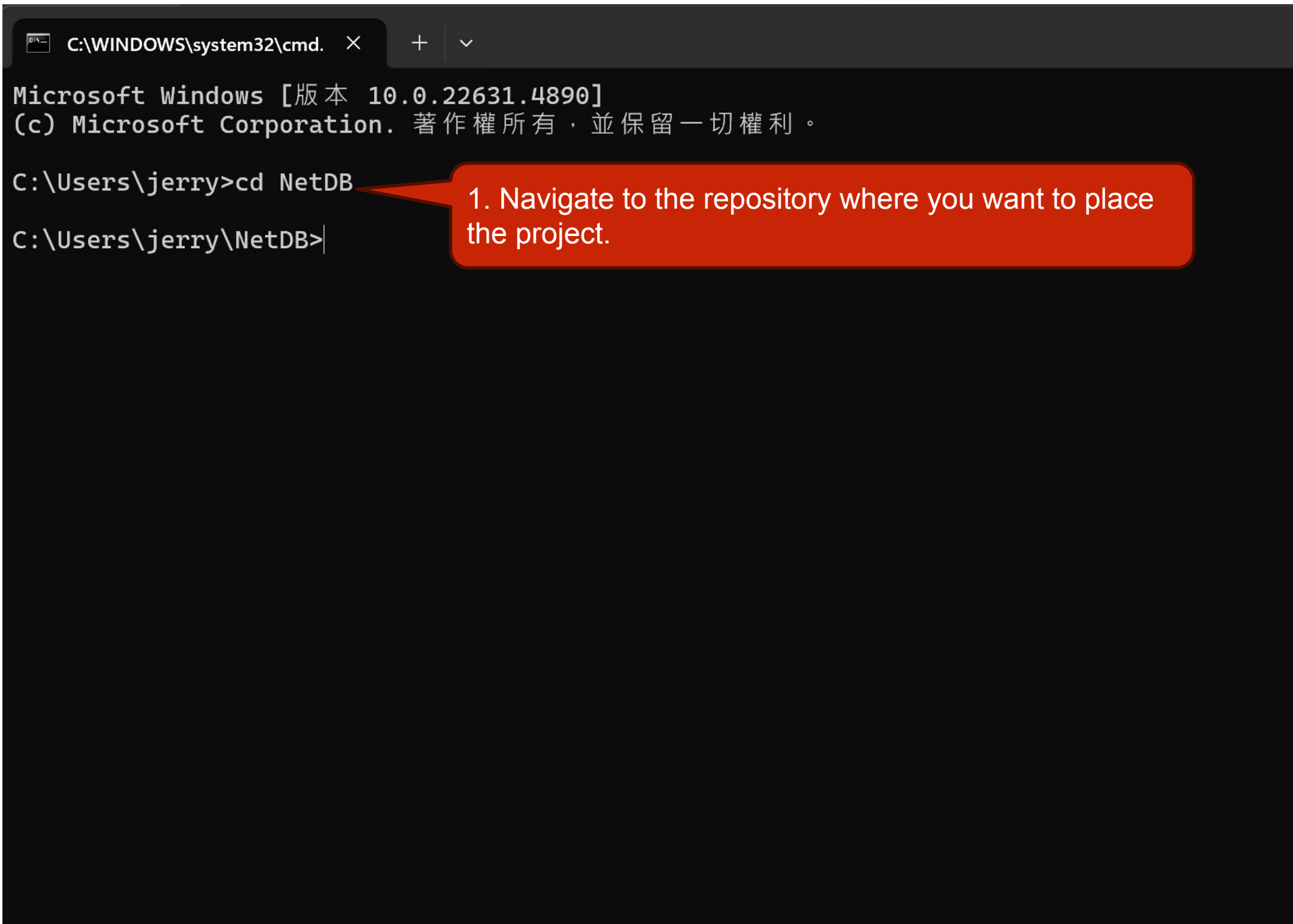


Basic Git Commands

- **git init**
 - Initialize a repository at current directory.
- **git add [file_name] (* git add . * → means add all files)**
 - Add files to git repository and let git track them.
- **git commit -m "commit messages"**
 - Save the changes to the git repository and create snapshots of the files.
- **git checkout [version]**
 - Go to a specific version.

scenario

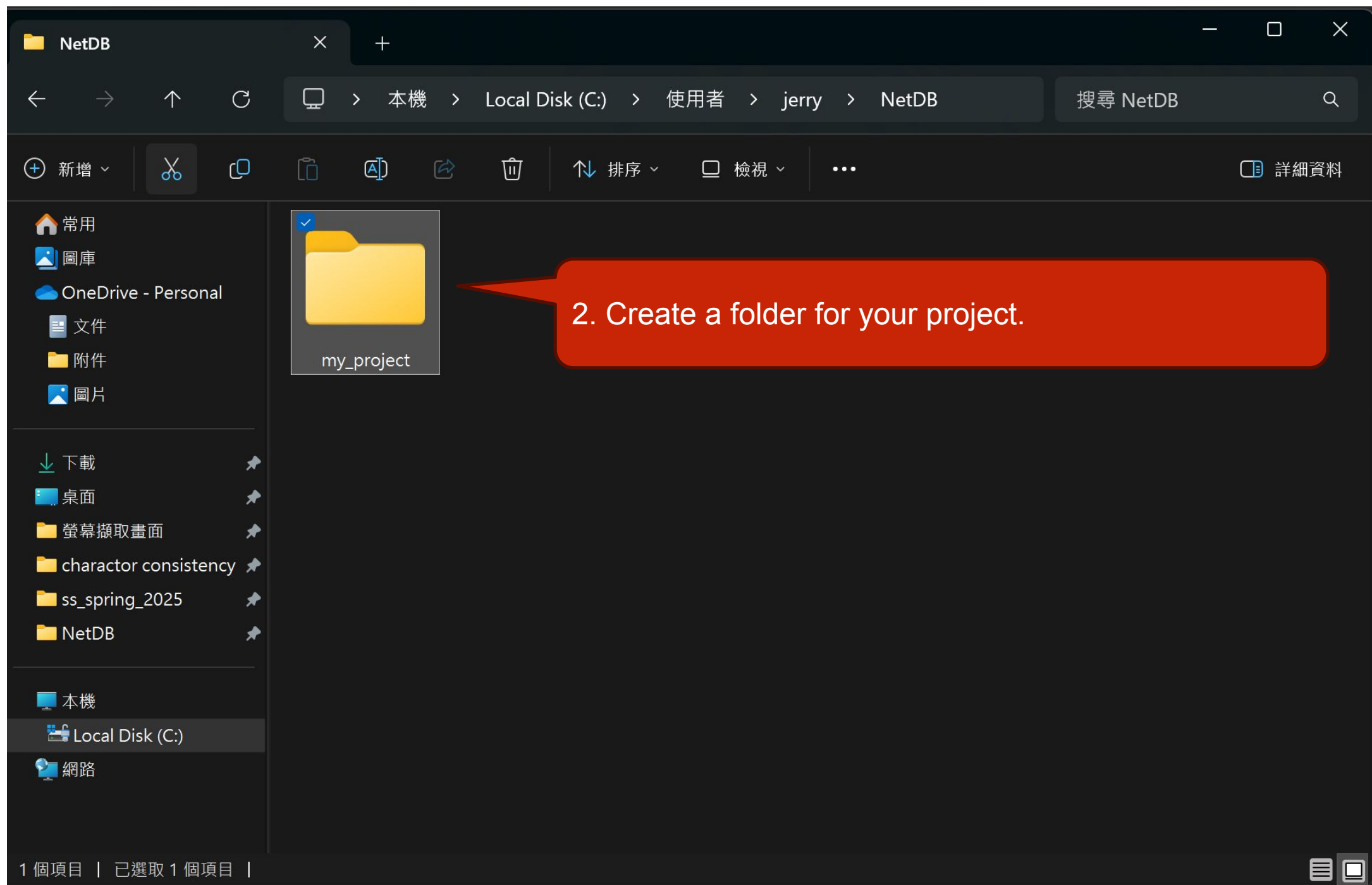
- . Create a project repository and initialize Git, and track the *HelloGit.txt* file for version control.

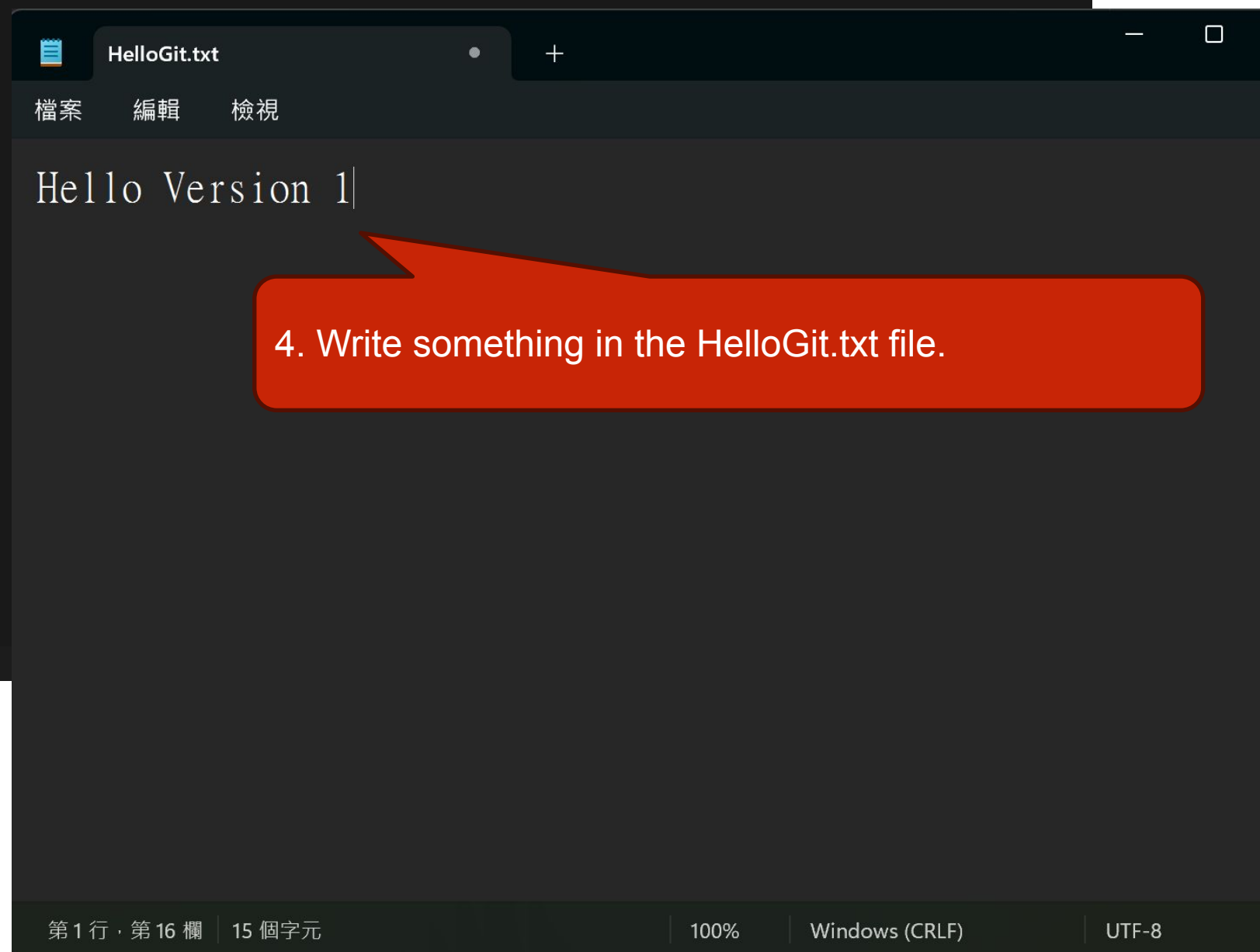
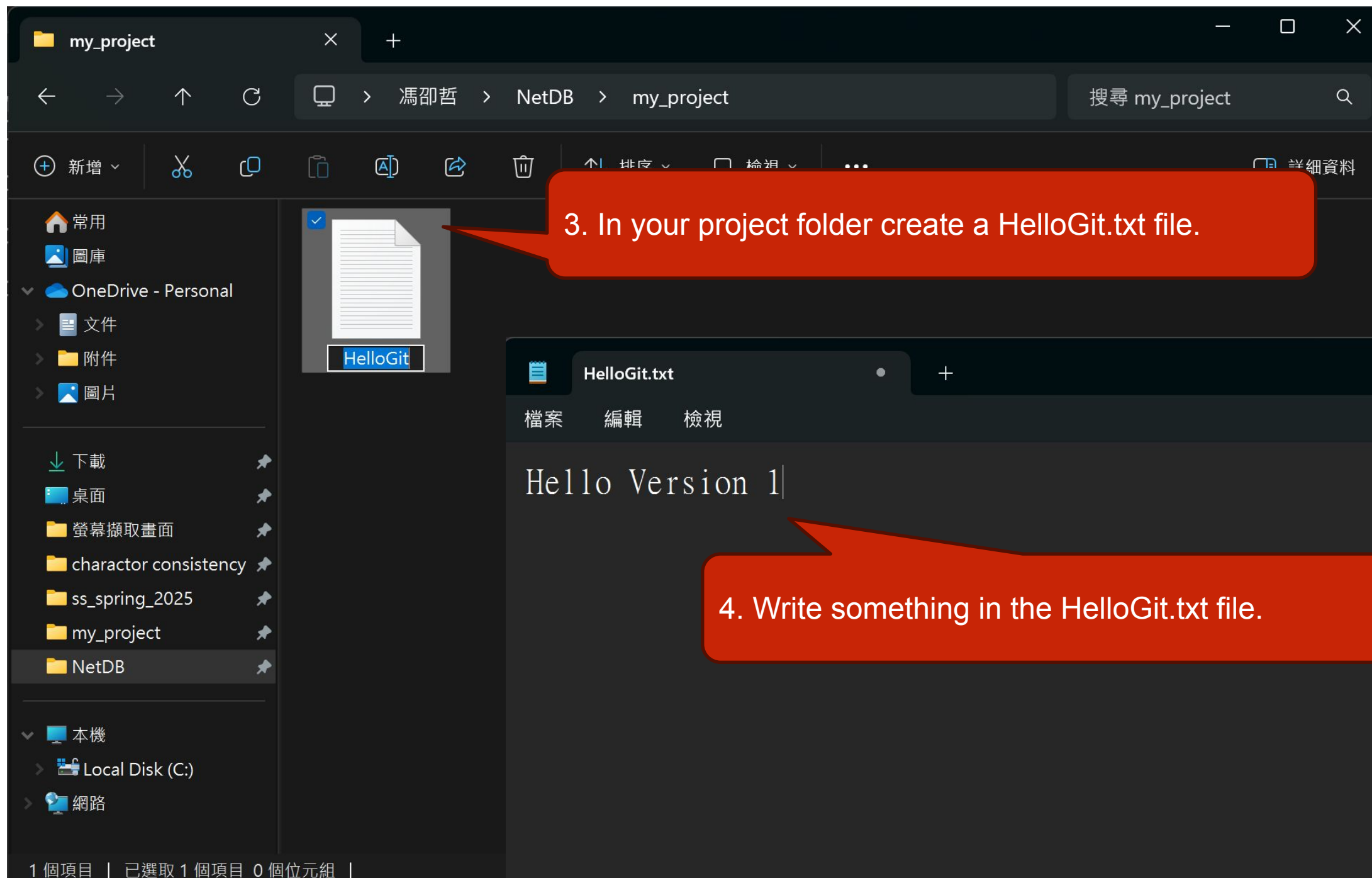


The image shows a Windows Command Prompt window with a dark background. The title bar at the top indicates the path 'C:\WINDOWS\system32\cmd.' and includes standard window controls. The command prompt displays the following text:

```
Microsoft Windows [版本 10.0.22631.4890]  
(c) Microsoft Corporation. 著作權所有，並保留一切權利。  
C:\Users\jerry>cd NetDB  
C:\Users\jerry\NetDB>|
```

An orange callout box with a pointer directed at the 'cd NetDB' command contains the text: '1. Navigate to the repository where you want to place the project.'





```
C:\WINDOWS\system32\cmd. × + v
Microsoft Windows [版本 10.0.22631.4890]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\jerry>cd NetDB
C:\Users\jerry\NetDB>cd my_project
C:\Users\jerry\NetDB\my_project>dir
磁碟區 C 中的磁碟是 Local Disk
磁碟區序號： 8697-6595

C:\Users\jerry\NetDB\my_project 的目錄

2025/02/18 下午 07:48 <DIR> .
2025/02/18 下午 07:47 <DIR> ..
2025/02/18 下午 07:49 15 HelloGit.txt.txt
1 個檔案 15 位元組
2 個目錄 190,638,534,656 位元組可用

C:\Users\jerry\NetDB\my_project>git init
Initialized empty Git repository in C:/Users/je
C:\Users\jerry\NetDB\my_project>
```

5. Navigate to the project fold.

6. Use the "dir" command to list the contents of a folder. (if windows)

7. Use the "git init" command to initialize a Git repository.

C:\WINDOWS\system32\cmd. X + v

Microsoft Windows [版本 10.0.22631.4890]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\jerry>cd NetDB

C:\Users\jerry\NetDB>cd my_project

C:\Users\jerry\NetDB\my_project>dir

磁碟區 C 中的磁碟是 Local Disk
磁碟區序號： 8697-6595

C:\Users\jerry\NetDB\my_project 的目錄

2025/02/18	下午 07:55	<DIR>	.
2025/02/18	下午 07:47	<DIR>	..
2025/02/18	下午 07:55		0 HelloGit.txt
	1 個檔案		0 位元組
	2 個目錄	190,626,844,672	位元組可用

C:\Users\jerry\NetDB\my_project>git init
Initialized empty Git repository in C:/Users/jerry/NetDB/my_project/.git/

C:\Users\jerry\NetDB\my_project>git add HelloGit.txt

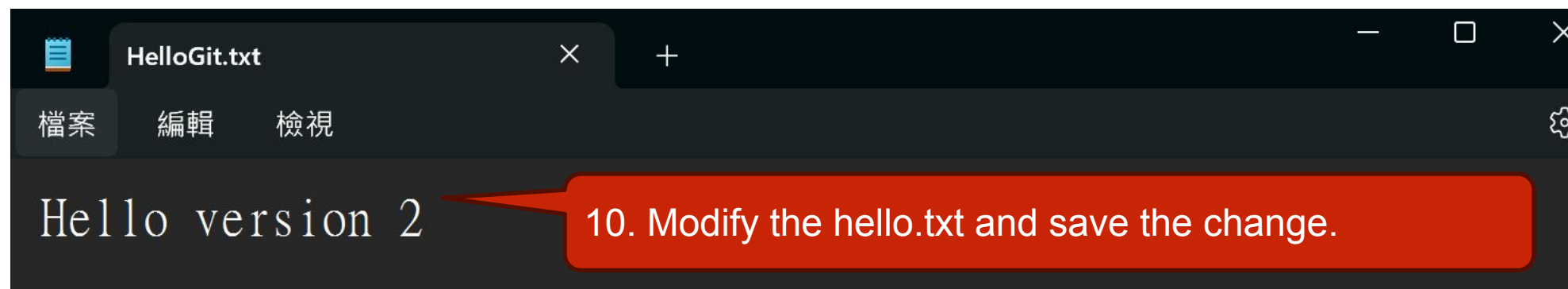
C:\Users\jerry\NetDB\my_project>git commit -m "version 1"

[master (root-commit) 3c0e49e] version 1
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 HelloGit.txt

C:\Users\jerry\NetDB\my_project>

8. Use the "git add" command to add HelloGit.txt to staging file.

9. Use the "git commit -m" command to commit the change.



```
C:\WINDOWS\system32\cmd. x + v

C:\Users\jerry\NetDB\my_project>dir
磁碟區 C 中的磁碟是 Local Disk
磁碟區序號： 8697-6595

C:\Users\jerry\NetDB\my_project 的目錄

2025/02/18 下午 08:03 <DIR> .
2025/02/18 下午 08:03 <DIR> ..
2025/02/18 下午 08:04 15 HelloGit.txt
                  15 位元組
                  1 個檔案
                  2 個目錄 190,627,364,864 位元組可用

C:\Users\jerry\NetDB\my_project>git init
Initialized empty Git repository in C:/Users/jerry/NetDB/my_project/.git/

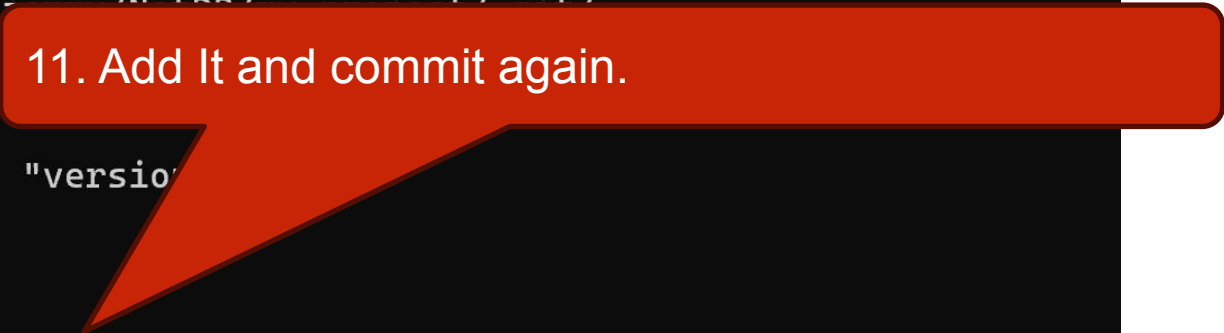
C:\Users\jerry\NetDB\my_project>git add HelloGit.txt

C:\Users\jerry\NetDB\my_project>git commit -m "version 1"
[master (root-commit) 7135d48] version 1
1 file changed, 1 insertion(+)
create mode 100644 HelloGit.txt

C:\Users\jerry\NetDB\my_project>git add HelloGit.txt

C:\Users\jerry\NetDB\my_project>git commit -m "version 2"
[master 7a9ac7f] version 2
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\jerry\NetDB\my_project>
```



第1行 · 第1欄

```
C:\WINDOWS\system32\cmd. x + v

C:\Users\jerry\NetDB\my_project>git init
Initialized empty Git repository in C:/Users/jerry/NetDB/my_project/.git/

C:\Users\jerry\NetDB\my_project>git add HelloGit.txt

C:\Users\jerry\NetDB\my_project>git commit -m "version 1"
[master (root-commit) 7135d48] version 1
1 file changed, 1 insertion(+)
create mode 100644 HelloGit.txt

C:\Users\jerry\NetDB\my_project>git

C:\Users\jerry\NetDB\my_project>git
[master 7a9ac7f] version 2
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\jerry\NetDB\my_project>git log
commit 7a9ac7fd7121170f61cca5e69d29ec423f987911 (HEAD -> master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:05:49 2025 +0800

    version 2

commit 7135d48a4ed19a2bb13a85408e1669ed8aa8340b
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:04:50 2025 +0800

    version 1

C:\Users\jerry\NetDB\my_project>
```

10. Use the "git log" command to view your versions.

Head:

A reference to where you are currently working on.

Master:

The default branch in a Git repository, typically where the final, stable version of the project is maintained.

C:\WINDOWS\system32\cmd. x + v

Switched to branch 'master'

C:\Users\jerry\NetDB\my_project>git log

commit 7a9ac7fd7121170f61cca5e69d29ec423f987911 (HEAD -> master)

Author: jerry <jerryfeng1103@gmail.com>

Date: Tue Feb 18 20:05:49 2025 +0800

version 2

Version ID

commit 7135d48a4ed19a2bb13a85408e1669ed8aa8340b

Author: jerry <jerryfeng1103@gmail.com>

Date: Tue Feb 18 20:04:50 2025 +0800

version 1

C:\Users\jerry\NetDB\my_project>git checkout 7135d48a4ed19a2bb13a85408e1669ed8aa8340b

Note: switching to '7135d48a4ed19a2bb13a85408e1669ed8aa8340b'.

You are in 'detached HEAD' state. You can look around, make experimental

changes and commit them, and you can discard any branches you are in this state without impacting any branches in your repository.

If you want to create a new branch to safely experiment with a new feature, do so (now or later) by using -

git switch -c <new-branch-name>

11. Go to specific version with the "git checkout" command and version ID

HelloGit.txt

檔案 編輯 檢視

Hello Version 1

12. Reopen the HelloGit.txt find yourself back to version 1 !


```
C:\WINDOWS\system32\cmd. X + v

commit 7a9ac7fd7121170f61cca5e69d29ec423f987911 [master]
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:05:49 2025 +0800

    version 2

commit 7135d48a4ed19a2bb13a85408e1669ed8aa8340b [HEAD]
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:04:50 2025 +0800

    version 1

C:\Users\jerry\NetDB\my_project>git checkout master
Previous HEAD position was 7135d48 version 1
Switched to branch 'master'

C:\Users\jerry\NetDB\my_project>git log
commit 7a9ac7fd7121170f61cca5e69d29ec423f987911 [HEAD -> master]
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:05:49 2025 +0800

    version 2

commit 7135d48a4ed19a2bb13a85408e1669ed8aa8340b
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:04:50 2025 +0800

    version 1

C:\Users\jerry\NetDB\my_project>
```

13. Use the “git checkout master” command to go back to the final version of your project

14. Reopen the HelloGit.txt again. Back to the latest version.

```
HelloGit.txt X +
檔案 編輯 檢視

Hello version 2

第 1 行 · 第 1 欄 15 個字元 100% Windows (CRLF)
```

LIFE IS
TOO SHORT
TO TYPE
THAT
VERSION ID.

which is 40 characters long...

```
C:\WINDOWS\system32\cmd. X + v

C:\Users\jerry\NetDB\my_project>git log
commit 7a9ac7fd7121170f61cca5e69d29ec423f987911 (HEAD -> master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:05:49 2025 +0800

    version 2

commit 7135d48a4ed19a2bb13a85408e1669ed8aa8340b
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 20:04:50 2025 +0800

    version 1

C:\Users\jerry\NetDB\my_project>git log --oneline
7a9ac7f (HEAD -> master) version 2
7135d48 version 1

C:\Users\jerry\NetDB\my_project>git checkout 7135d48
Note: switching to '7135d48'.

You are in 'detached HEAD' state. You can look at, make
changes and commit them, and you can discard any
state without impacting any branches by switching
back.

If you want to create a new branch to retain
do so (now or later) by using -c with the switch
command.

git switch -c <new-branch-name>
```

15. Use the "git log --oneline" command to get the shorter version ID.

16. Use shorter version ID to check specific version.

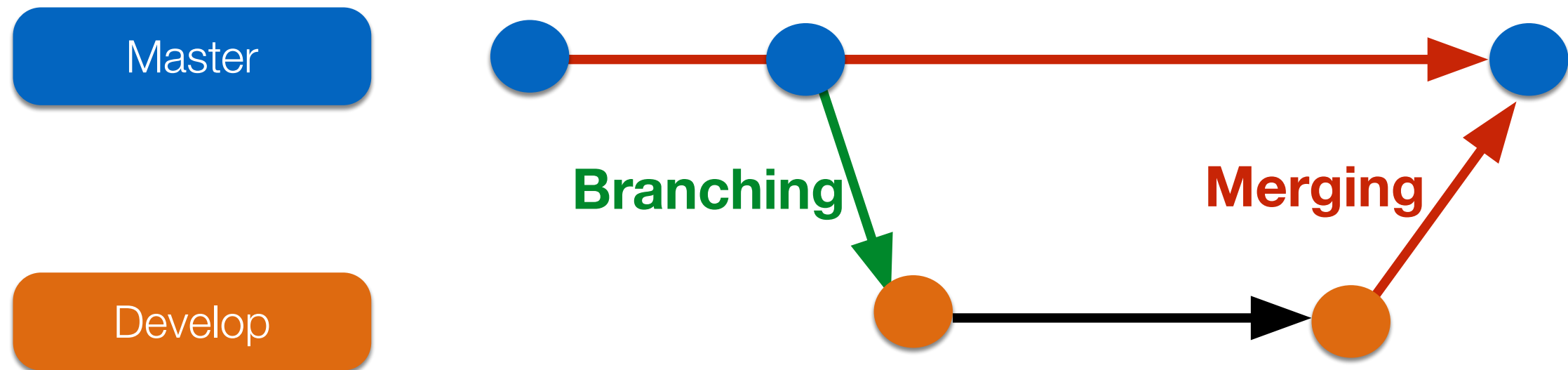
```
HelloGit.txt
檔案 編輯 檢視

Hello Version 1|
```

Outline

- General Rule
- Introduction to Git
- Version control
- Branch and merge
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Git Branch



Git commands (1/2)

- Branching steps
 - Creating a new branch

```
git branch [branch name]
```

- Checking out the branch

```
git checkout [branch name]
```

Git commands (2/2)

- Merging steps
 - Checking out a branch to merge

```
git checkout [branch 1 name]
```

- Merging another branch

```
git merge [branch 2 name]
```

scenario

- Assume you are working on a new feature in a Git repository and want to create a separate branch work on it and then merge it back into master branch

-

The image shows a Windows file explorer window titled "my_project" with a sidebar on the left. The main pane displays a folder named ".git" and a file named "file". A red callout bubble points to the ".git" folder with the text: "1. Initially there is a file.txt in ~\.NetDB\my_project with git already initialized".

Overlaid on the file explorer is a text editor window titled "file.txt" showing the text: "Hello, this is master branch." This text is enclosed in a red box, and a red label "Initial content in file.txt" points to it.

Below the text editor is a command prompt window titled "C:\WINDOWS\system32\cmd." showing the following commands and output:

```
C:\Users\jerry>cd NetDB
C:\Users\jerry\NetDB>cd my_project
C:\Users\jerry\NetDB\my_project>git log
commit 8a3db010cd8ae5f682d81ede54b1c647f5f44bb9 (HEAD -> master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 22:47:39 2025 +0800

Initial commit on master brach
```

The output of the "git log" command is enclosed in a red box, and a red label "Initial log of master branch" points to it.

At the bottom of the command prompt window, the text "[無標題]" is visible.

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [版本 10.0.22631.4890]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\jerry>cd NetDB

C:\Users\jerry\NetDB>cd my_project

C:\Users\jerry\NetDB\my_project>git log
commit 8a3db010cd8ae5f682d81ede54b1c647f5f44bb9 (HEAD ->
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 22:47:39 2025 +0800

    Initial commit on master brach

C:\Users\jerry\NetDB\my_project>git branch feature-branch

C:\Users\jerry\NetDB\my_project>git checkout feature-branch
Switched to branch 'feature-branch'

C:\Users\jerry\NetDB\my_project>
```

2. Create a new Branch using the “git branch [branch name] ” command

3. Switch to the new branch using the “git checkout [branch name] ” command

file.txt

檔案 編輯 檢視

Hello, this is the master branch.
This is a new feature.

4. Make Changes in the feature branch

C:\WINDOWS\system32\cmd.

Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:01:19 2025 +0800

Initial commit on master branch

C:\Users\jerry\NetDB\my_project>git branch feature-branch

C:\Users\jerry\NetDB\my_project>git checkout feature-branch
Switched to branch 'feature-branch'

C:\Users\jerry\NetDB\my_project>git add *

C:\Users\jerry\NetDB\my_project>git commit -m "Added a new feature"
[feature-branch dfd304e] Added a new feature
1 file changed, 1 insertion(+)

C:\Users\jerry\NetDB\my_project>git log
commit dfd304ed1b565bf1c21f3c0a4e9f7cfd0b7a8ff8 (HEAD -> feature-branch)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:03:46 2025 +0800

Added a new feature

commit fb903f40d1c918951ca48ce41347c805ab203a21 (master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:01:19 2025 +0800

Initial commit on master branch

C:\Users\jerry\NetDB\my_project>

5. Add and commit the check in feature branch. (use "git add *" to stage all files in the current directory)

Since we already switch to feature branch. "git log" will show the log in feature branch

file.txt

檔案 編輯 檢視

Hello, this is the master branch.
This is a new feature.
This is a second feature.

6. Perform another change, repeat the previous steps, and observe the Git log.

```
C:\WINDOWS\system32\cmd. x + v
Date: Tue Feb 18 23:01:19 2025 +0800
Initial commit on master branch
C:\Users\jerry\NetDB\my_project>git add *
C:\Users\jerry\NetDB\my_project>git commit -m "Addend a second feature"
[feature-branch 22d4ba2] Addend a second feature
1 file changed, 2 insertions(+), 1 deletion(-)
C:\Users\jerry\NetDB\my_project>git log
commit 22d4ba229447145c0565807a898433f7bf8cbb6e (HEAD -> feature-branch)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:09:08 2025 +0800
Addend a second feature
commit dfd304ed1b565bf1c21f3c0a4e9f7cfd0b7a8ff8
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:03:46 2025 +0800
Added a new feature
commit fb903f40d1c918951ca48ce41347c805ab203a21 (master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:01:19 2025 +0800
Initial commit on master branch
C:\Users\jerry\NetDB\my_project>
```

```
C:\WINDOWS\system32\cmd. X + v  
C:\Users\jerry\NetDB\my_project>git checkout master  
Switched to branch 'master'  
C:\Users\jerry\NetDB\my_project>git log  
commit fb903f40d1c918951ca48ce41347c805ab203a21 (HEAD -> master)  
Author: jerry <jerryfeng1103@gmail.com>  
Date: Tue Feb 18 23:01:19 2025 +0800  
  
Initial commit on master branch  
C:\Users\jerry\NetDB\my_project>
```

7. Switch Back to master branch

8. The log remains unchanged in the master branch."

```
file.txt +  
檔案 編輯 檢視  
Hello, this is the master branch.
```

9. Reopen the file.txt and the content go back to initial

The image shows a code editor window with a tab labeled 'file.txt'. The editor contains the text 'Hello, this is the master branch.' and 'How are you?'. The text 'How are you?' is highlighted with a red box. Below the editor is a terminal window showing the following commands and output:

```
C:\Users\jerry\NetDB\my_project>git log
commit fb903f40d1c918951ca48ce41347c805ab203a21 (HEAD -> master)
Author: jerry <jerryfeng1103@gmail.com>
Date: Tue Feb 18 23:01:19 2025 +0800

    Initial commit on master branch

C:\Users\jerry\NetDB\my_project>git add *

C:\Users\jerry\NetDB\my_project>git commit -m "add content to master branch"
[master e3363b1] add content to master branch
1 file changed, 1 insertion(+)
```

A red callout box points to the terminal with the text: '10. Try to also make change in master branch and commit the change'. Another red box highlights the command 'git commit -m "add content to master branch"' in the terminal.

```
C:\WINDOWS\system32\cmd. X + v  
C:\Users\jerry\NetDB\my_project>git log master --oneline  
e3363b1 (HEAD -> master) add content to master branch  
fb903f4 Initial commit on master branch  
  
C:\Users\jerry\NetDB\my_project>git log feature-branch --oneline  
22d4ba2 (feature-branch) Addend a second feature  
dfd304e Added a new feature  
fb903f4 Initial commit on master branch  
  
C:\Users\jerry\NetDB\my_project>
```

As observed in the log, a commit in one branch will not affect another branch.

C:\WINDOWS\system32\cmd.

C:\Users\jerry\NetDB\my_project>git checkout master

Already on 'master'

C:\Users\jerry\NetDB\my_project>git merge feature-branch

Auto-merging file.txt

CONFLICT (content): Merge conflict in file.txt

Automatic merge failed; fix conflicts and then commit the result.

C:\Users\jerry\NetDB\my_project>

11. Ensure to move back to master branch (or any target branch you want to merge with)

12. Use the “git merge [branch name]” to merge the branch

Note that there is a merge conflict

file.txt

檔案 編輯 檢視

Hello, this is the master branch.

<<<<<<< HEAD

How are you?

=====

This is a new feature.

This is a second feature.

>>>>>>> feature-branch

13. Reopen the file.txt file then you found that there is merge conflict mark

<<<<<<< HEAD:

Show the change from your target branch.

=====:

Separator between the conflicting change

>>>>>>> feature-branch

The changes from the branch you are trying to marge with the target branch

47



file.txt



檔案

編輯

檢視

Hello, this is the master branch
How are you?
This is a new feature.
This is a second feature.

14. Decide which version to keep and edit the file to resolve a merge conflict and save the change. In this case I reserve all the version.



C:\WINDOWS\system32\cmd.



Already on 'master'

```
C:\Users\jerry\NetDB\my_project>git merge feature-branch
```

```
Auto-merging file.txt
```

```
CONFLICT (content): Merge conflict in file.txt
```

```
Automatic merge failed; fix conflicts and then commit
```

```
C:\Users\jerry\NetDB\my_project>git add *
```

```
C:\Users\jerry\NetDB\my_project>git commit -m "Resolved merge conflict in file.txt"
```

```
[master 2aed399] Resolved merge conflict in file.txt
```

```
C:\Users\jerry\NetDB\my_project>
```

15. Add and commit again

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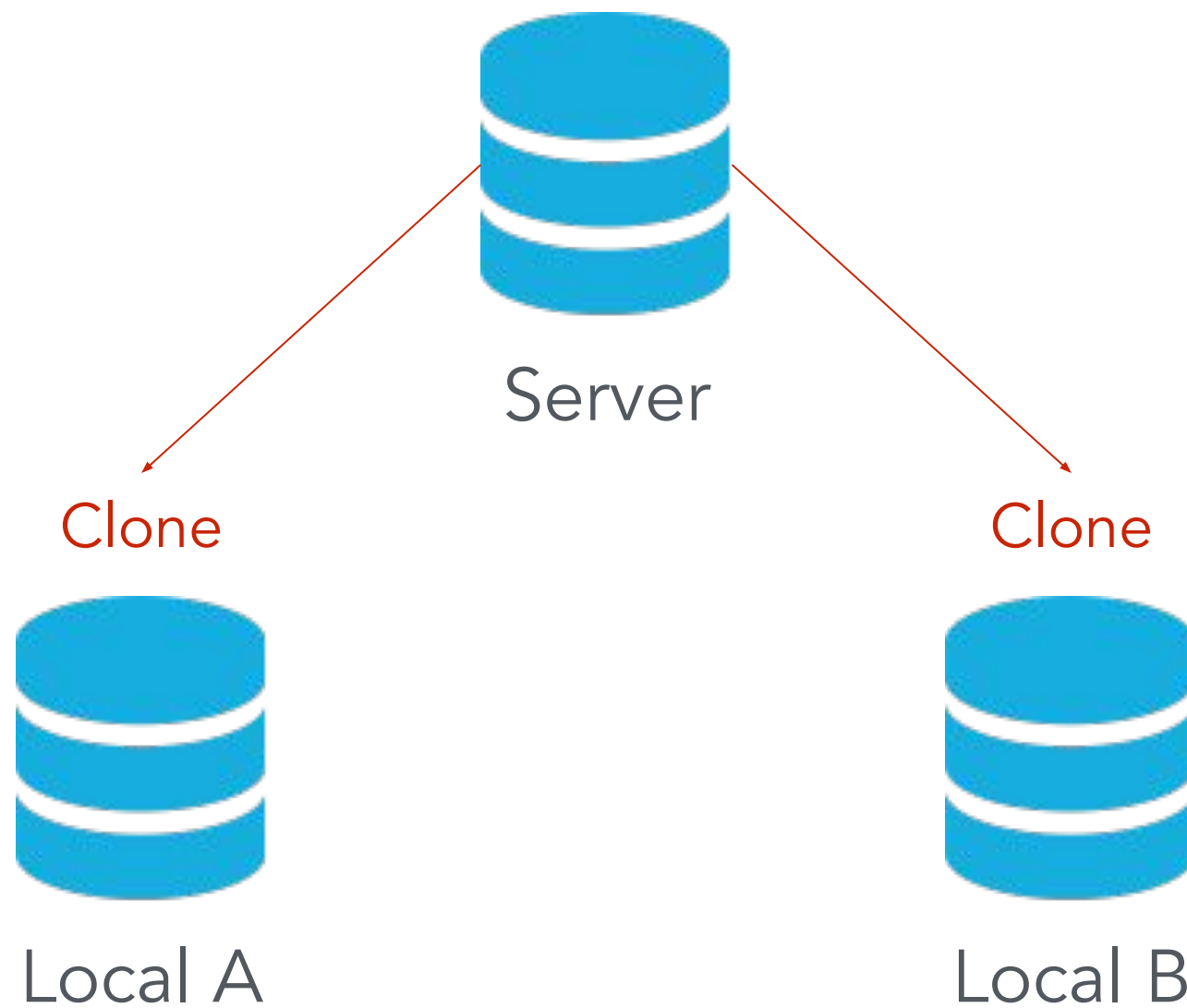
Collaboration

- To work with others using git, you'll need a server that store the repository.
- Git is distributed, which means
 - Everyone can store a copy of the repository downloaded from the server to their computer and do their jobs independently.

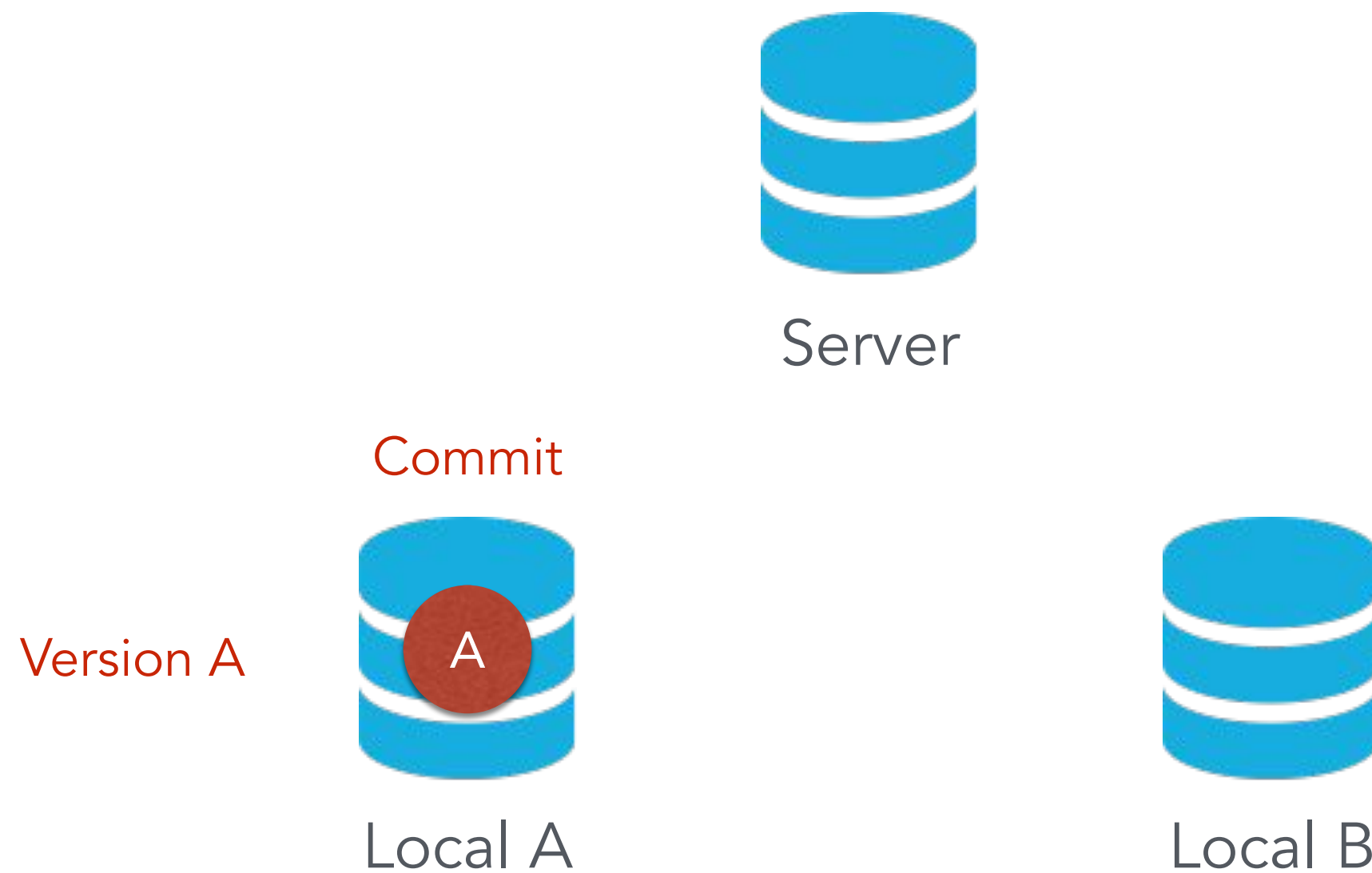
Collaboration Workflows (1/2)

1. If you don't have the project, *clone* (download) the repository from the server.
2. Do your work and commit the changes at local. Once done, *push* (upload) the repository to the server.
3. If someone else modified the project, you can *pull* (sync) the repository to get the updated project.
4. Repeat 2 and 3.

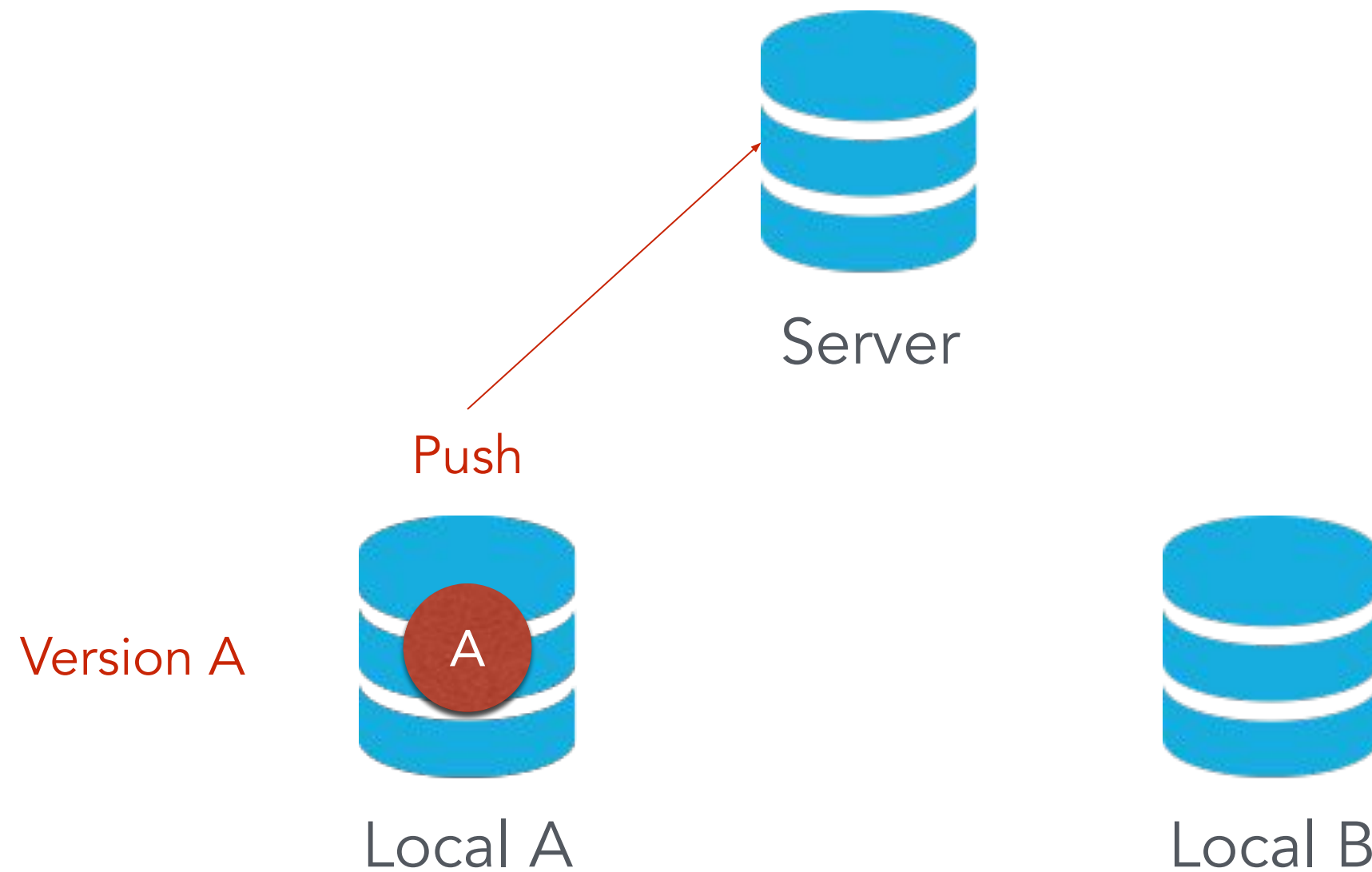
Collaboration Workflows (2/2)



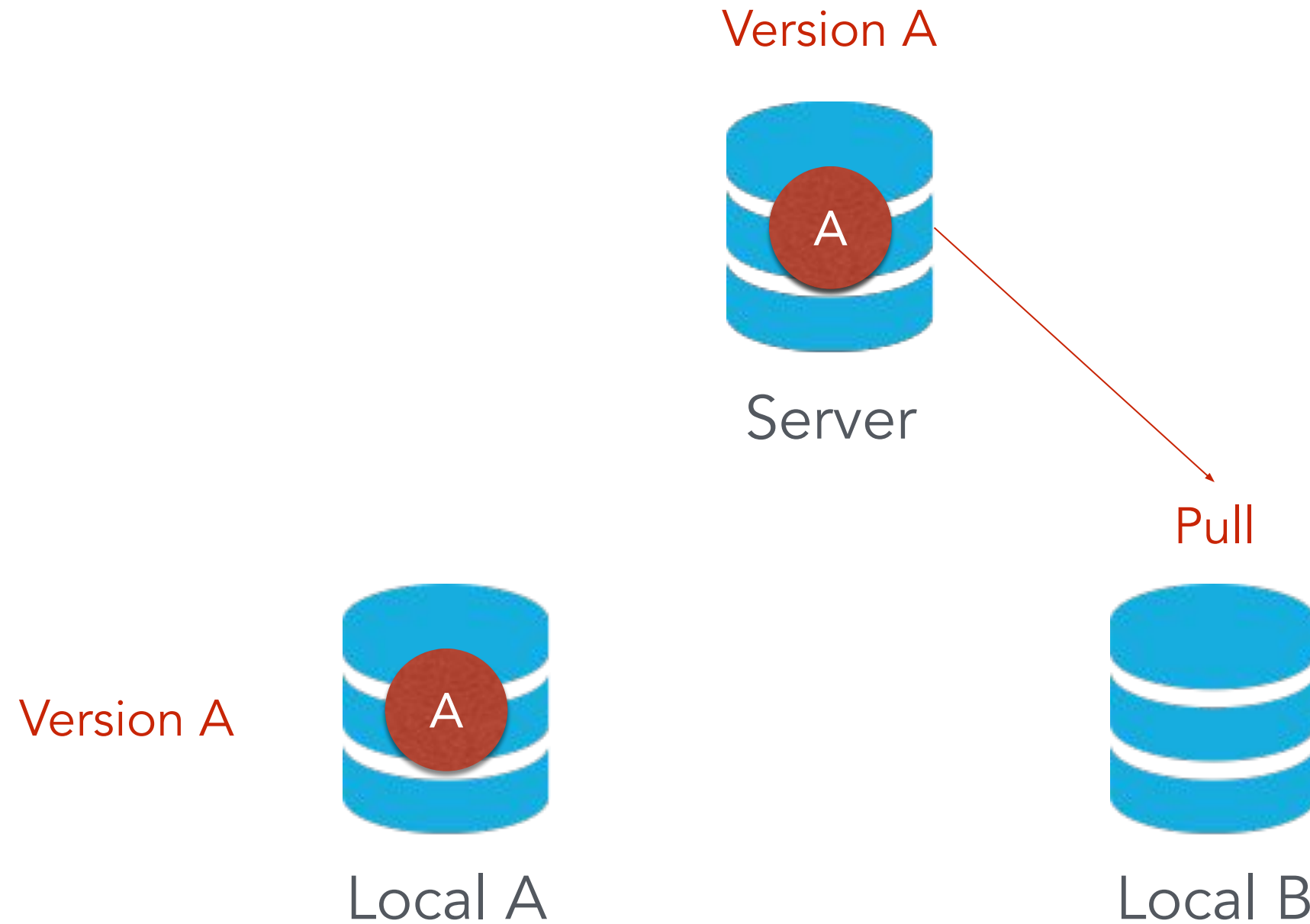
Collaboration Workflows (2/2)



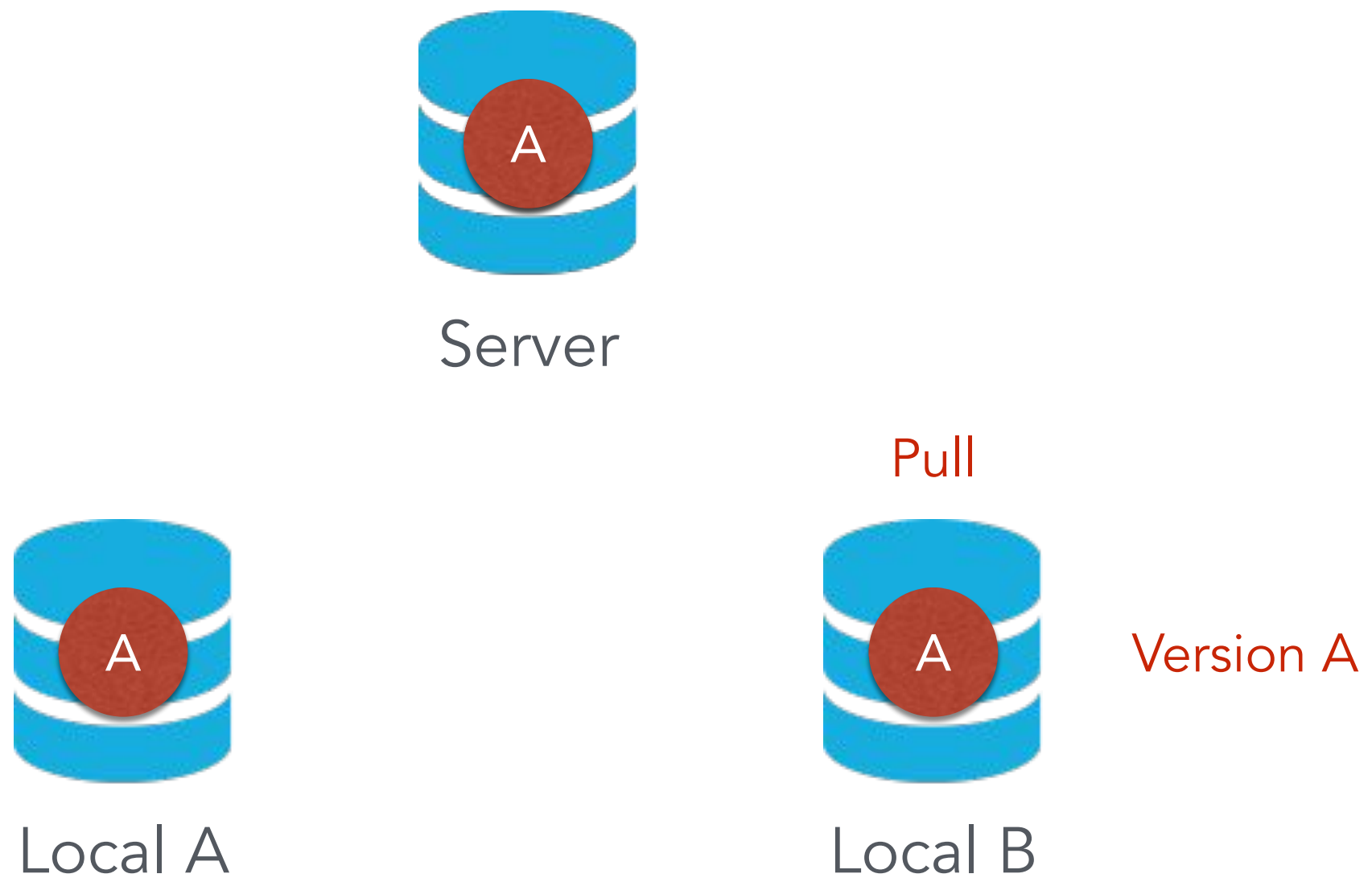
Collaboration Workflows (2/2)



Collaboration Workflows (2/2)

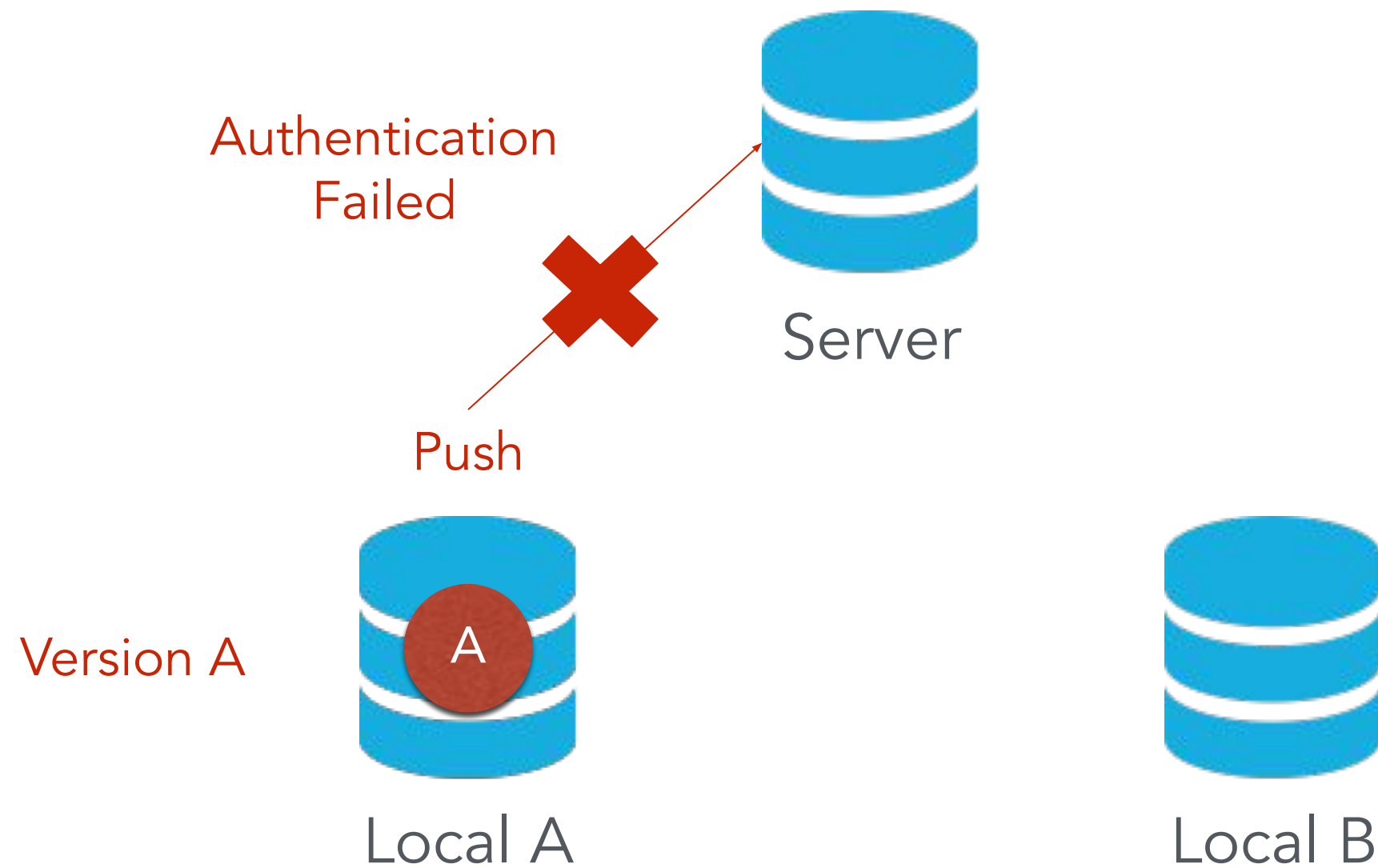


Collaboration Workflows (2/2)



Something went wrong.

Authentication Failed



Why Authentication Failed?

Collaboration Workflow

- If you tried to clone the code template from a server and want to **push** the modified file.
 - You will get authentication failed.
 - It's because it was a **project of others**, which means you are not able to save the changes back to the server.
- So, how can I copy a project from others on a open source platform like Github?

Introducing
Fork



Author

Original
Project



Forked
Project



Forked from Red



Commit



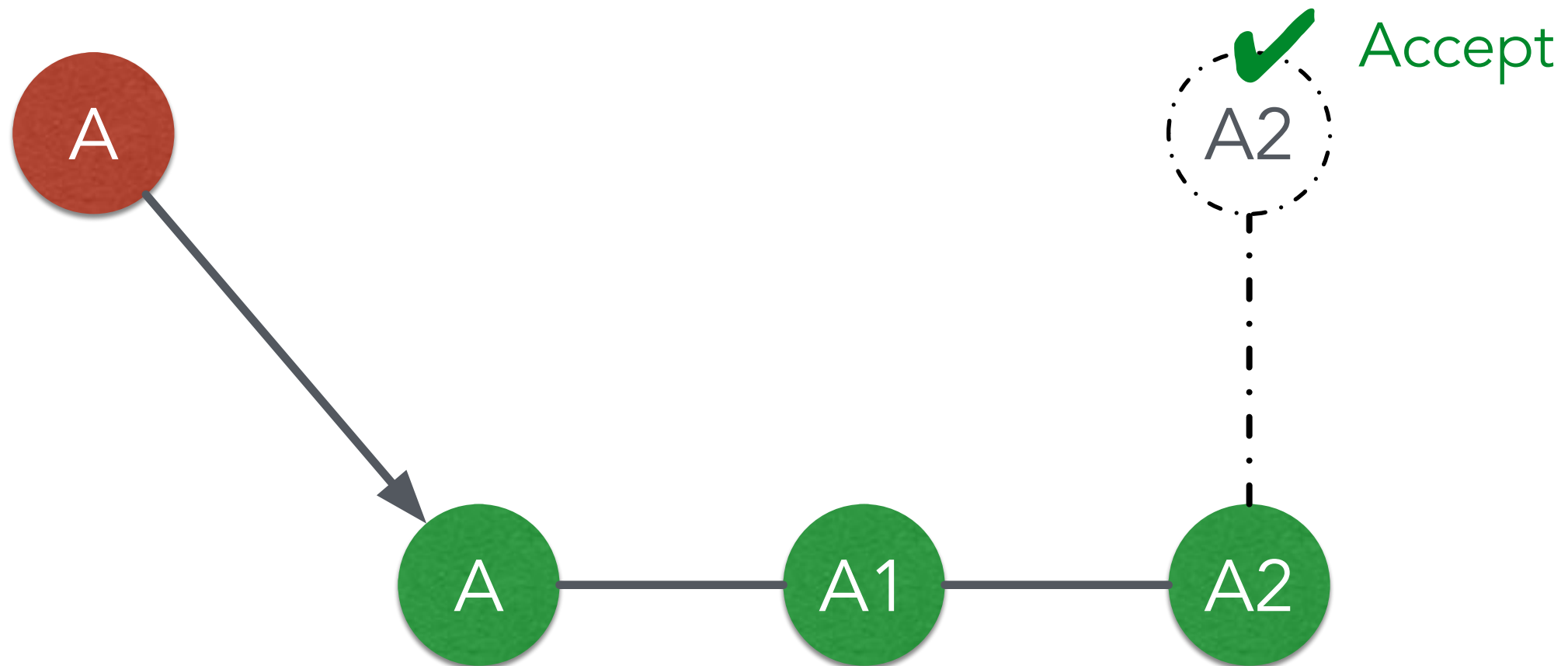
Commit



Open a Merge Request

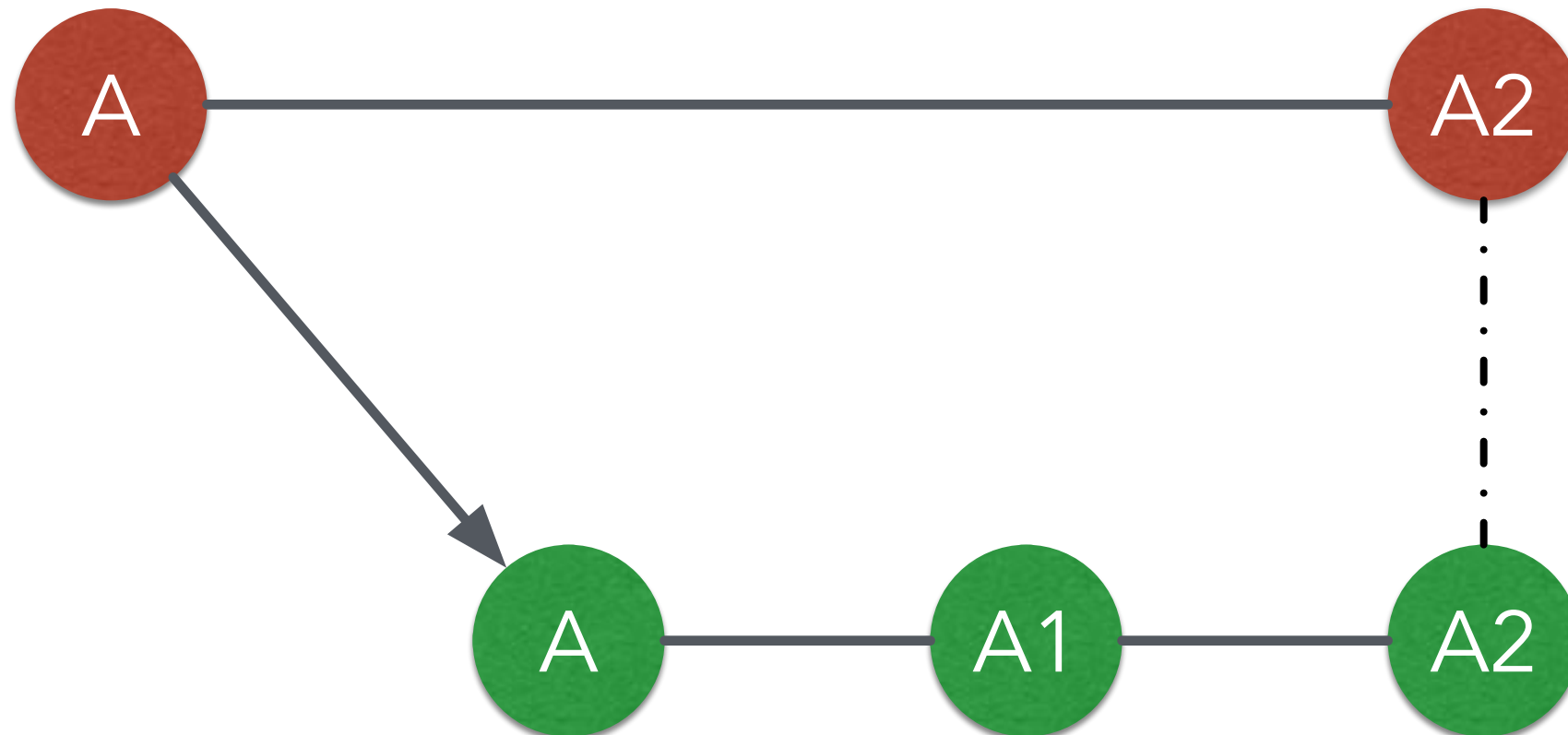
Editor

Author

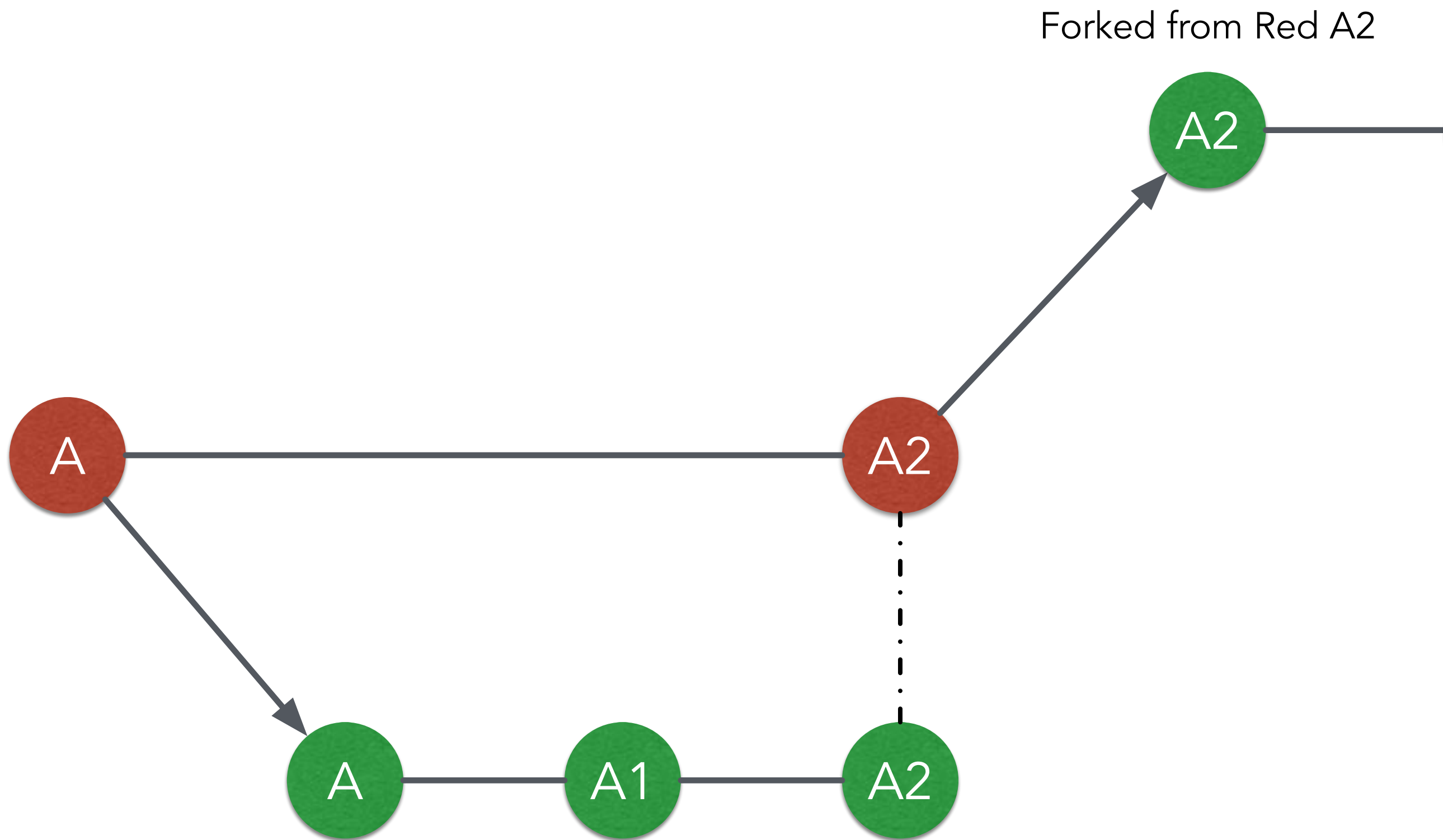


Editor

Author



Editor



Git Collaboration Workflow

1. *Fork* a repository to make a copy of it.
2. *Clone* the repository you forked to your workspace.
3. Do your work and *commit* the changes in your workspace.
4. *Push* the repository to the server to synchronize them.
5. Open a *merge request* to origin repository .

Basic Git Commands (2/2)

- `git clone [url]`
 - Clone a repository from remote server
- `git push [url] [branch-name]`
 - Push committed file to remote server

Outline

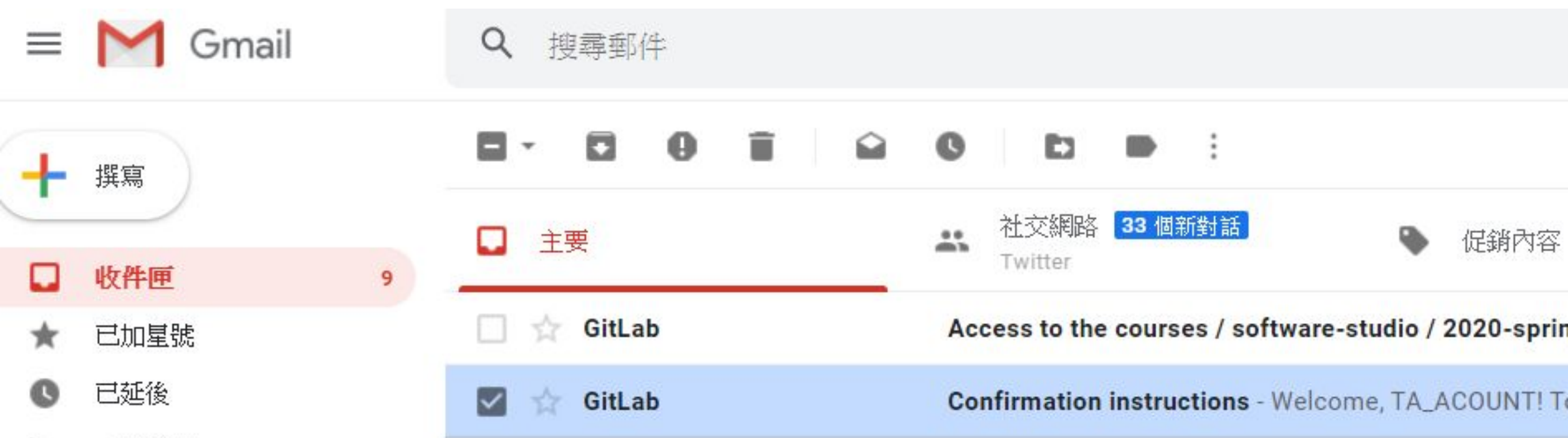
- General Rule
- Introduction to Git
 - Version control
 - Git Basics
 - Try Git!
 - Remote Repositories
- How to Submit Your Code to Gitlab
- Tools & References



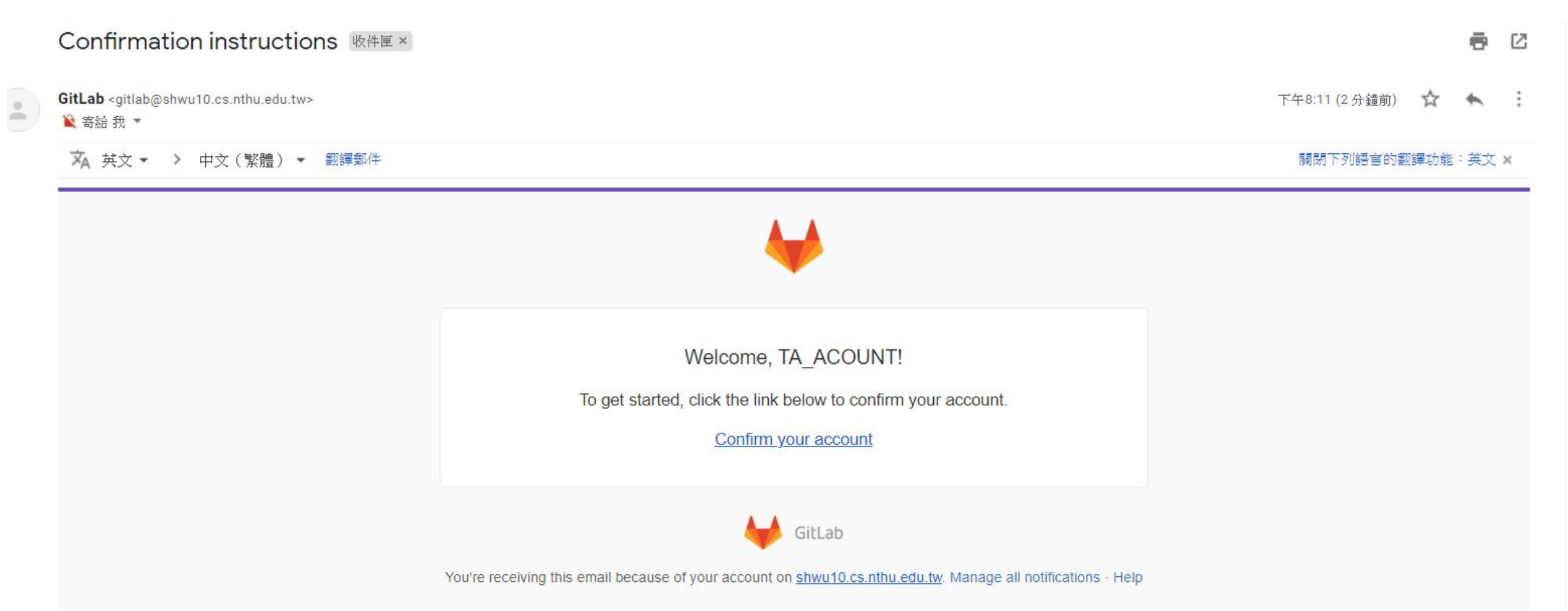
Gitlab

- We have created account for you
- Account: Email (e.g. ssta2026@datalab.cs.nthu.edu.tw)
- Password: student ID (e.g. 114012345)

Gitlab



Gitlab



Gitlab



Your email address has been successfully confirmed. Please sign in.

DataLab

Welcome to GitLab for DataLab.

Email: your email
Password: studentID

Sign in

Username or email

Password

☐ Remember me

[Forgot your password?](#)

Sign in

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository

You can access course projects in [this group](#)

courses >  > 2025-spring

2

2025-spring 

Group ID: 6011 



New subgroup

New project

Subgroups and projects

Shared projects

Archived projects

Search by name

Name



S

submission-exercise 

★ 0

1 hour ago

 **submission-exercise** 

Project ID: 8735 

   Star 0  Fork 0

2 Commits 1 Branch 0 Tags 41 KB Files 41 KB Storage

master submission-exercise / +

History Find file Web IDE  Clone



Update README.md

Shao-Che Feng authored 1 hour ago

19a597ef



 Upload File

 README

 Auto DevOps enabled

 Add LICENSE

 Add CHANGELOG

 Add CONTRIBUTING

 Configure Integrations

Name	Last commit	Last update
 README.md	Update README.md	1 hour ago

 README.md

Lab 1 - Practice Submission

This repository is built for practicing submissions for assignments and projects. You can follow the instructions below in order to know the whole workflow for submitting a lab or project.

Try it!

1. Fork this project.
2. Clone the **forked** project from Gitlab to your local environment.
3. Add a new file and write something.
4. Commit your work.
5. Push the repository to the server.

1. Click to fork



Fork project

A fork is a copy of a project.
Forking a repository allows you to make changes without affecting the original project.

Project name

submission-exercise

Project URL

https://shwu10.cs.nthu.edu.tw/

Select a namespace

Project slug

submission-exercise

Want to house several dependent projects u

2. Select your name

Project description (optional)

Visibility level ?

☒ Private

Project access must be granted explicitly to each user. If this project is part of a group, access will be granted to members of the group.

☐ Internal

The project can be accessed by members of the group.

☐ Public

The project can be accessed without any authentication.

3. Click Private

Fork project

Cancel

4. Click to fork

S submission-exercise

- Project information
- Repository
- Issues 0
- Merge requests 0
- Security & Compliance
- Deployments
- Monitor
- Infrastructure
- Packages & Registries
- Analytics
- Wiki
- Snippets
- Settings

Chen Yu-Hsuan > submission-exercise

5. Check if this repository is under your account

Project ID: 3798

3 Commits 99 Branches 0 Tags 102 KB Files 113 KB Storage

Forked from [courses / software-studio / 2023-spring / submission-exercise](#)

master

submission-exercise /



History

Find file

Web IDE



Clone



finish lab 1

Chen Yu-Hsuan authored 1 year ago

1654ad5c



Upload File

README

Auto DevOps enabled

Add LICENSE

Add CHANGELOG

Add CONTRIBUTING

Configure Integrations

Name

Last commit

Last update

README.md

finish lab 1

1 year ago

6. Go to settings

GitLab Menu

S

submission-exercise

Project information

Repository

Issues0

Merge requests0

Security & Compliance

Deployments

Monitor

Infrastructure

Packages & Registries

Analytics

Wiki

Snippets

Settings

General

Integrations

Webhooks

Access Tokens

Repository

Monitor

Packages & Registries

Usage Quotas

Max file size is 200 KB.

Save changes

Visibility, project features, permissions

Collapse

Choose visibility level, enable/disable project features and their permissions, disable email notifications, and show default award emoji.

Project visibility

Manage who can see the project in the public access directory. [Learn more.](#)

Private

The project is accessible only by members of the project. Access must be granted explicitly to each user.

Issues

Flexible tool to collaboratively develop ideas and plan work in this project. [Learn more.](#)

Only Project Members

Repository

View and edit files in this project.

Only Project Members

Merge requests

Submit changes to be merged upstream.

Only Project Members

Forks

Users can copy the repository to a new project.

Only Project Members

7. Set project to private

 **GitLab** Menu

S

submission-exercise

 Project information

 Repository

 Issues

0

 Merge requests

0

 Security & Compliance

 Deployments

 Monitor

 Infrastructure

 Packages & Registries

 Analytics

 Wiki

 Snippets

 Settings

General

Integrations

Webhooks

Only Project Members

☐ **Disable email notifications**

Override user notification preferences for all project members.

☒ **Show default award emojis**

Always show thumbs-up and thumbs-down award emoji buttons on issues, merge requests, and snippets.

☒ **Warn about high severity merge requests**

High severity merge requests are merge requests that are marked as high severity. They are used in many languages, but can also be used in potential merge requests.

Save changes

Merge requests

Choose your merge method, merge options, merge checks, and merge suggestions.

Expand

Badges

Customize this project's badges. [What are badges?](#)

Expand

Service Desk

Expand

8. Scroll down and save changes

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository



submission-exercise

Project ID: 6444



☆ Star 0

 Fork 0

0 Commits

1 Branch

0 Tags

20 KB Files

20 KB Storage

master

submission-exercise /

+

History

Find file

Web IDE

↓

Clone



Initial commit

Chen Yu-Hsuan authored 3 days ago

Upload File

Auto DevOps enabled

Add README

Add LICENSE

Add

Configure Integrations

Name	Last commit
 README.md	Initial commit

 README.md

Lab 1 - Practice Submission

This repository is built for practicing submissions for assignments and projects. You can follow the instructions below in order to know the whole workflow for submitting a lab or project.

Try it!

1. Fork this project.
2. Clone the **forked** project from Gitlab to your local environment.
3. Add a new file and write something.
4. Commit your work.
5. Push the repository to the server.
6. Send a merge request of **your branch** to the origin repository.

Clone with SSH

git@shwu10.cs.nthu.edu.tw:

Clone with HTTPS

https://shwu10.cs.nthu.edu.tw/cour

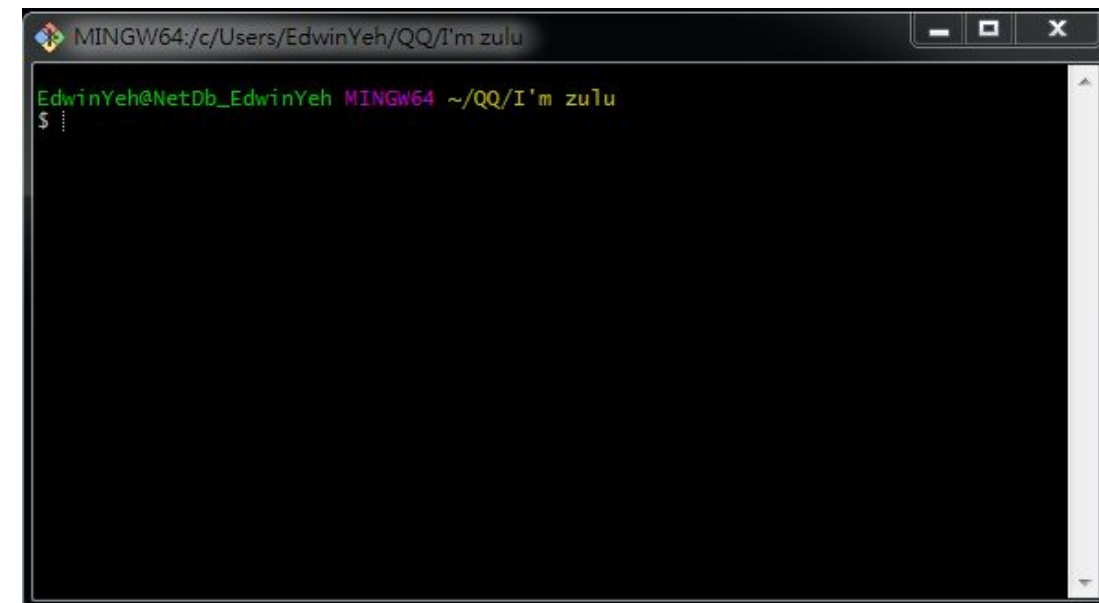
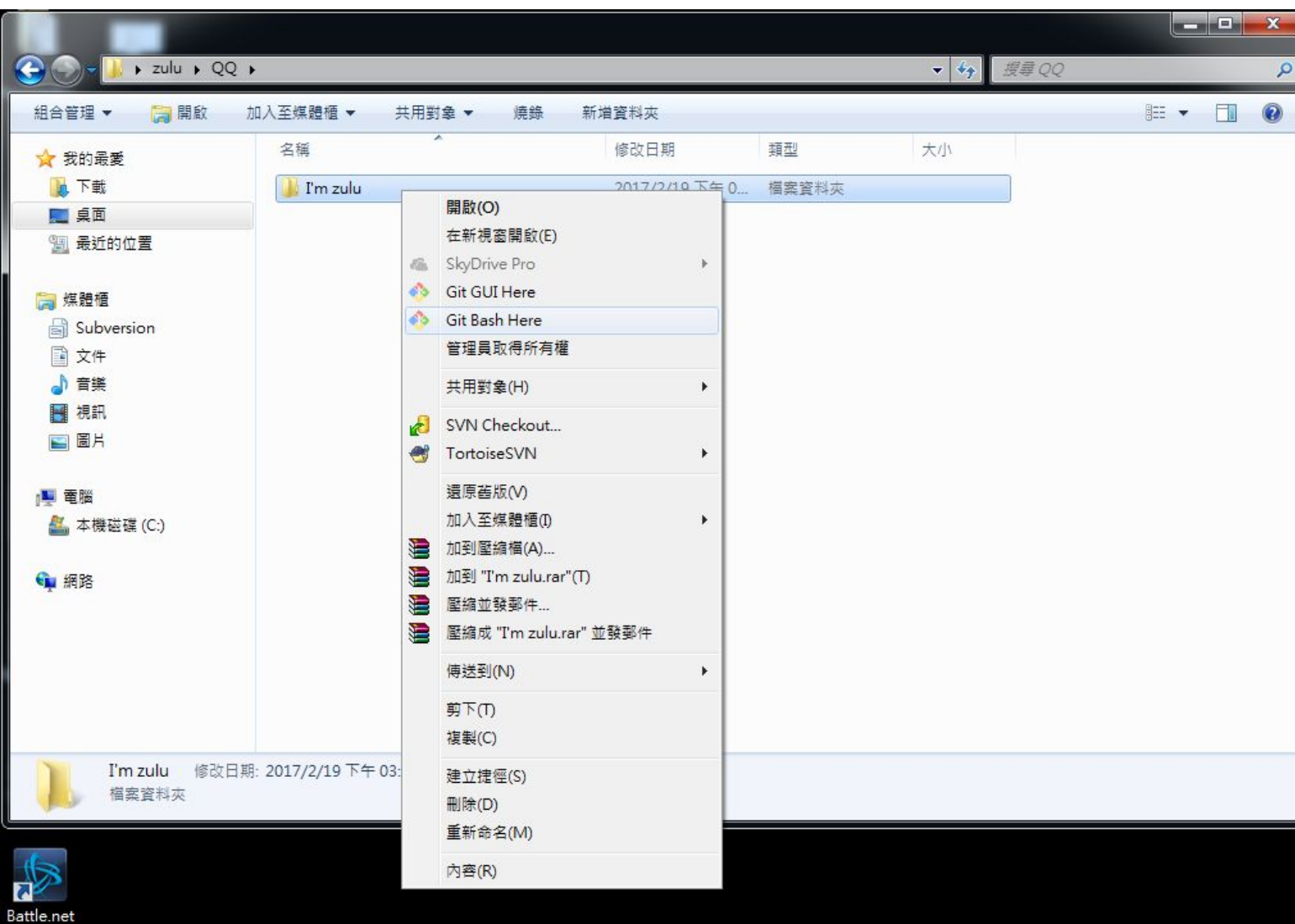
Visual Studio Code (SSH)

Visual Studio Code (HTTPS)

1. Choose HTTPS

2. Copy the link

If You use Windows



3. Create a folder to put your repos

```
~/SS-Projects ➤ git clone https://shwu10.cs.nthu.edu.tw/ss-student/submission-exercise.git
Cloning into 'submission-exercise'...
remote: Counting objects: 3, done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0)
Unpacking objects: 100% (3/3), done.
~/SS-Projects ➤ ls
submission-exercise
~/SS-Projects ➤
```

4. Type "git clone {URL}"

5. The repo has been successfully cloned

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository

```
~/SS-Projects/submission-exercise master vim lab1.js
~/SS-Projects/submission-exercise master git add -A
~/SS-Projects/submission-exercise master + git status
```

1. -A means all files

```
On branch master
Your branch is up-to-date with 'origin/master'.
Changes to be committed:
  (use "git reset HEAD <file>..." to unstage)

    new file:   lab1.js
```

2. Check if your file is added to git

```
~/SS-Projects/submission-exercise master + git commit -m "Finish lab1"
[master c1acaf4] Finish lab1
1 file changed, 1 insertion(+)
create mode 100644 lab1.js
~/SS-Projects/submission-exercise master
```

3. Commit your changes

```
~/SS-Projects/submission-exercise master vim lab1.html
~/SS-Projects/submission-exercise master git add -A
~/SS-Projects/submission-exercise master + git commit -m "Finish lab1"
```

```
[master 8a603d9] Finish lab1
```

```
Committer: Real Wei <realwei@Realweis-MBP.local>
```

Your name and email address were configured automatically based on your username and hostname. Please check that they are accurate. You can suppress this message by setting them explicitly:

```
git config --global user.name "Your Name"
git config --global user.email you@example.com
```

After doing this, you may fix the identity used for this commit with

```
git commit --amend --reset-author
```

```
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 lab1.html
```

```
~/SS-Projects/submission-exercise master
```

If you see these message, type
`git config --global user.name "{name}"`
`git config --global user.email "{email}"`

{email} is the email you use on gitlab

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository

```
~/SS-Projects/submission-exercise ? master git push -u origin master
```

```
Counting objects: 6, done.
```

```
Delta compression using up to 4 threads.
```

```
Compressing objects: 100% (4/4), done.
```

```
Writing objects: 100% (6/6), 497 bytes | 0 bytes/s, done.
```

```
Total 6 (delta 1), reused 0 (delta 0)
```

```
To https://shwu10.cs.nthu.edu.tw/ss-student/submission-exercise.git
```

```
b1e0571..8a603d9 master -> master
```

```
Branch master set up to track remote branch master from origin.
```

```
~/SS-Projects/submission-exercise ? master
```

Type "git push -u origin master"

Workflow

- For each lab, you should follow the workflow below
 1. Fork our template repository on Gitlab
 2. Clone the **forked** repository to your computer
 3. Finish your lab
 4. Commit in your computer
 5. Push to Gitlab
 6. Send merge request of **your branch** to our template repository

S submission-exercise

Project information

Repository

Issues0

Merge requests0

Security & Compliance

Deployments

Monitor

Infrastructure

Packages & Registries

Analytics

Wiki

Snippets

Settings

Chen Yu-Hsuan > submission-exercise

S submission-exercise

Project ID: 3798

Star0

Fork0

1. Click Merge Requests

KB Files 113 KB Storage

Forked from courses / software-studio / 2023-spring / submission-exercise

master submission-exercise / +

History

Find file

Web IDE

Clone

finish lab 1

Chen Yu-Hsuan authored 1 year ago

1654ad5c

Upload File

README

Auto DevOps enabled

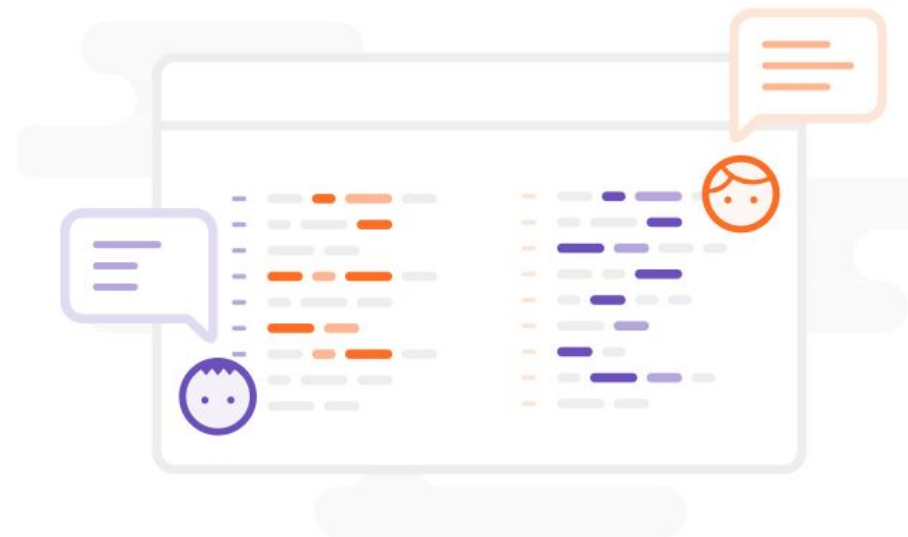
Add LICENSE

Add CHANGELOG

Add CONTRIBUTING

Configure Integrations

Name	Last commit	Last update
M+ README.md	finish lab 1	1 year ago



Merge requests are a place to propose changes you've made to a project and discuss those changes with others

Interested parties can even contribute by pushing commits if they want to.

New merge request

New merge request

2. New merge request

GitLab Menu

submission-exercise-2

Project information

Repository

Issues 0

Merge requests 0

Security & Compliance

Deployments

Monitor

Infrastructure

Packages & Registries

Analytics

Wiki

Snippets

Settings

Chen Yu-Hsuan > submission-exercise-2

New merge request

Source branch

yhch/submission-exercise-2

master

Initial commit

Chen Yu-Hsuan authored 3 days ago

21779350

Compare branches and continue

Target branch

courses/software-studio/2025-ss/subm...

master

Select target branch

00000000

No matching results

3. Choose the branch you pushed in your repo

Ensure that the commit in the version record of the selected branch matches the commit name of the final version you have in mind.

 GitLab  Menu

submission-exercise-2

- Project information
- Repository
- Issues 0
- Merge requests 0
- Security & Compliance
- Deployments
- Monitor
- Infrastructure
- Packages & Registries
- Analytics
- Wiki
- Snippets
- Settings

Chen Yu-Hsuan > submission-exercise-2 > Merge requests > New

New merge request

Source branch

yhch/submission-exercise-2 master

 Initial commit
Chen Yu-Hsuan authored 3 days ago

21779350 

Compare branches and continue

Target branch

courses/software-studio/2025-ss/subm... master

Select target branch

00000000 

No matching results

4. Choose the branch named after your ID

Ensure that the source of the selected branch is the class folder (courses/software-studio/2025-spring/submission-exercise) and your student ID (not master or someone else's student ID).

 GitLab

Menu

+

Search GitLab

Q











submission-exercise-2

Project information

Repository

Issues0

Merge requests0

Security & Compliance

Deployments

Monitor

Infrastructure

Packages & Registries

Analytics

Wiki

Snippets

Settings

Chen Yu-Hsuan > submission-exercise-2 > Merge requests > New

New merge request

Source branch

yhch/submission-exercise-2

master

 Initial commit
Chen Yu-Hsuan authored 3 days ago

21779350



Compare branches and continue

Target branch

courses/software-studio/2025-ss/subm...

master

Select target branch

00000000

No matching results

5. Compare branches

You can also confirm or change whether the selected branch is correct in this section. For instance, in the picture, the course branch selected is master, which is incorrect. The correct one should be `courses/software-studio/2025-spring/submission-exercise: {your student ID}`.

GitLab Menu

Chen Yu-Hsuan > submission-exercise-2 > Merge requests > New

New merge request

From `yhch/submission-exercise-2:master` into `Courses/software-studio/2025-spring/submission-exercise: {ID}`

6. Set title to "{ID} Submission"

Title

Start the title with `Draft:` to prevent a merge request draft from merging before it's ready.
Add [description templates](#) to help your contributors to communicate effectively!

Description

Write Preview

Describe the goal of the changes and what reviewers should be aware of.

Markdown and quick actions are supported

[Attach a file](#)

Description

Write Preview

B *I* ≡ `</>` [🔗](#) [≡](#) [≡](#) [≡](#) [📄](#) [📄](#) [↗](#)

Describe the goal of the changes and what reviewers should be aware of.

Markdown and quick actions are supported

[📎 Attach a file](#)

Assignee

Unassigned



[Assign to me](#)

Reviewer

Unassigned



Milestone

Milestone



Labels

Labels



Merge options

☐ Squash commits when merge request is accepted. [?](#)

Contribution

☐ Allow commits from members who can merge to the target branch. [About this feature](#)

Not available for private projects

Create merge request



7. If everything is OK,
submit your merge request

Notice - Don't do this

A search bar with the text 'gitlab' entered. On the right side of the bar is a small microphone icon.

Google 搜尋

好手氣

Notice – Don't do this



約有 8,300,000 項結果 (搜尋時間：0.45 秒)

GitLab

<https://gitlab.com/> ▼ 翻譯這個網頁

這項網站搜尋結果說明因為網站的 robots.txt 而無法提供瞭解詳情

GitLab.com | GitLab

<https://about.gitlab.com/gitlab-com/> ▼ 翻譯這個網頁

GitLab.com. unlimited free repositories and collaborators. Sign Up. Free public & private repositories and unlimited collaborators. Runs GitLab Enterprise Edition ...

GitLab介紹— Practical guide for git users 0.1 文档

git-tutorial.readthedocs.io/zh/latest/gitlab.html ▼

GitLab介紹¶. 目前最流行的線上Git專案管理系統可以說是非GitHub 莫屬，對於一般OpenSource的專案選擇使用GitHub做為線上Git專案管理系統即可，也免收任何 ...

GitHub - gitlabhq/gitlabhq: GitLab CE | Please open new issues in our ...

<https://github.com/gitlabhq/gitlabhq> ▼ 翻譯這個網頁

README.md. GitLab. Build status CE coverage report Code Climate Core Infrastructure Initiative Best Practices. Canonical source. The canonical source of ...

Gitlab - 維基百科，自由的百科全書 - Wikipedia

<https://zh.wikipedia.org/zh-tw/Gitlab> ▼

GitLab是一個利用Ruby on Rails開發的開源應用程式，實現一個自代管的Git專案倉庫，可通過Web介面進行存取公開的或者私人專案。它擁有與GitHub類似的功能， ...

Notice - Don't do this



GitLab.com

GitLab.com offers free unlimited (private) repositories and unlimited collaborators.

- [Explore projects on GitLab.com](#) (no login needed)
- [More information about GitLab.com](#)
- [GitLab.com Support Forum](#)

By signing up for and by signing in to this service you accept our:

- [Privacy policy](#)
- [GitLab.com Terms.](#)

Sign in	Register
Username or email	
Password	
☐ Remember me	Forgot your password?
Sign in	

Didn't receive a confirmation email? [Request a new one.](#)

Sign in with



The GitLab setup in our laboratory is accessible through the link.

Shan-Hung Wu

Description

Announcement

Curriculum ▾

Resources

Resources

Here are some course materials and resources related to this course. For code and its details (such as assigned reading, project links, quiz, etc.) please refer to the GitLab. For online forum please refer to the iLMS system.



Here!!!!

Outline

- General Rule
- Introduction to Git
- Version control
- Branch and merge
- How to Submit Your Code to Gitlab
- Tools & References

Tools

- Git GUI
 - GitKraken
- Editor / IDE
 - Visual Studio Code
 - Atom
 - Sublime Text
 - Brackets
 - Notepad++
 - Webstorm



axosoft

GitKraken

Viewing 112/151 Show All

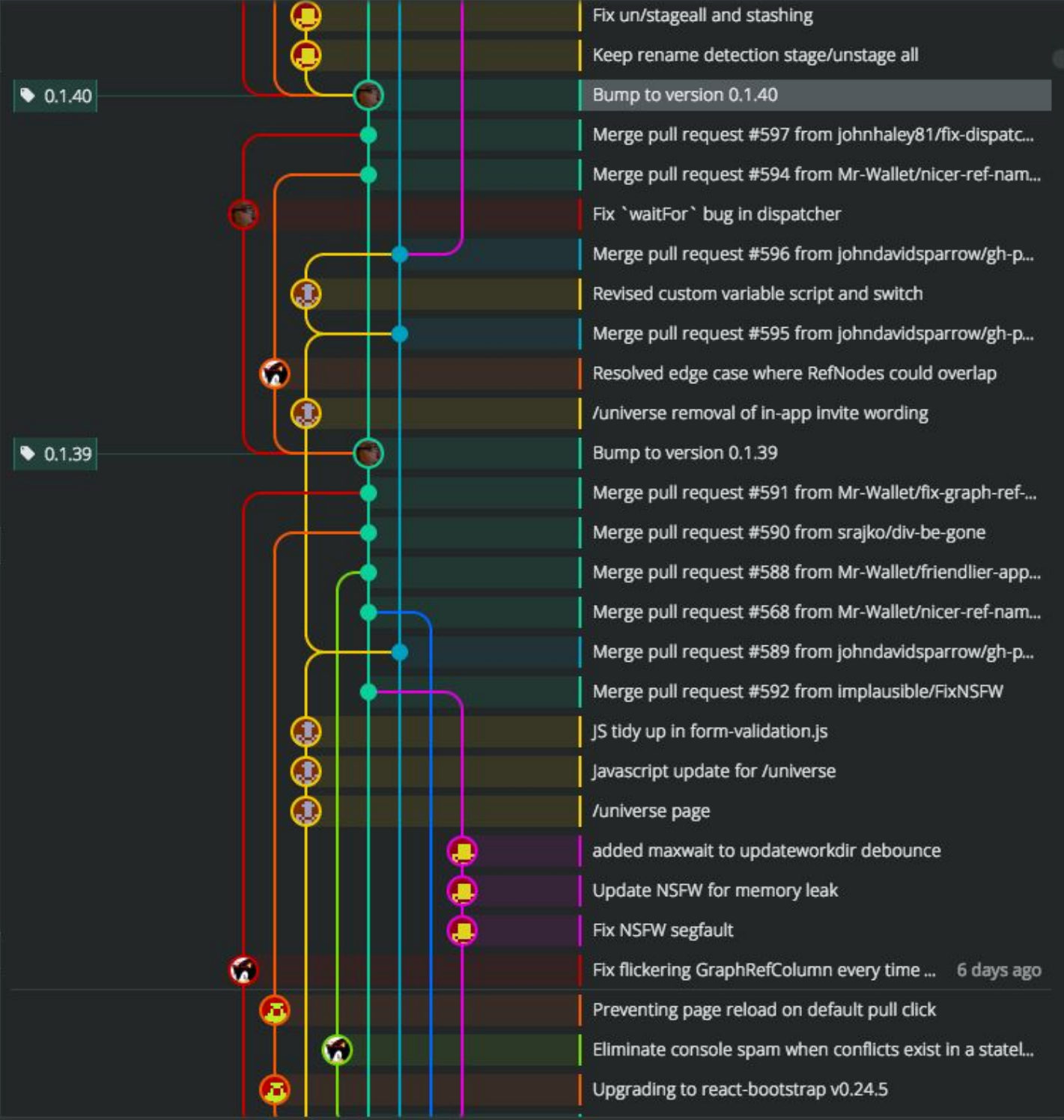
LOCAL (7/11)

- fancier-refbar-changes
- fancy-responsive-refbar-it... 42 ↗ 99+
- graph-color-test
- hopscotch 24 ↗ 99+
- init-repo-gitignore-typeahead
- invite-system 6 ↗ 99+
- jars-view-file-history
- master 5 ↙
- remote-panel-redesign 15 ↗ 13
- settings-theme-styling
- view-file-history 24 ↗ 99+

REMOTE (6/41)

- Jeff-Schinella (0/1)
- Jordan-Wallet (0/7)
- Justin-GK (0/1)
- Ken-Price (0/2)
- Kyle-Smith (2/8)
- Max-Korp (0/2)
- Sjegan-Rajko (0/8)
- ayresa (0/3)
- cbargren (0/5)
- origin (4/4)

TAGS (99/99)



Commit: cca151e6b9e32c3f9209c25131706740050
Parent: 8efe30a11761983173f844900fa5ec5c6be2
Author: John Haley <johnh@axosoft.com>
Author Date: September 30th 2015, 2:54 pm

Bump to version 0.1.40

0 added 0 deleted 2 modified

npm-shrinkwrap.json

```
@@ -1,6 +1,6 @@
1 1 {
2 2   "name": "gitkraken",
-3 2   "version": "0.1.39",
+3 3   "version": "0.1.40",
4 4   "dependencies": {
5 5     "atom-keymap": {
6 6       "version": "5.1.11",
```

package.json

```
@@ -1,7 +1,7 @@
1 1 {
2 2   "name": "gitkraken",
3 3   "productName": "GitKraken",
-4 3   "version": "0.1.39",
+4 4   "version": "0.1.40",
5 5   "description": "An intuitive git cli
6 6   "main": "./src/appBootstrap/main.js"
7 7
```

Provide Feedback



VS Code

EXPLORE

WORKING FILES

03.jpg

TBL-STYLES

css

img

js

hoverIntent.js

jquery.dropdown.js

jquery.more.js

jquery.more.min.js

jquery.plugin.js

jquery.plugin.min.js

mapper.js

maputil.js

navigation.js

smoothscroll.js

tabs.js

navigation.js

```

1 var scriptbase = _spPageContextInfo.webServerRelativeUrl + "/_layouts/15/";
2
3 $(document).ready(function () {
4     $.getScript(scriptbase + "SP.Runtime.js", function () {
5         $.getScript(scriptbase + "SP.js", function () {
6             $.getScript(scriptbase + "SP.Taxonomy.js", function () {
7                 context = SP.ClientContext.get_current();
8                 //Call your code here.
9                 console.log("Navigation - ready to rock.");
10
11                 // Get default termstore
12
13                 session = SP.Taxonomy.TaxonomySession.getTaxonomySession(context);
14                 termStore = session.getDefaultSiteCollectionTermStore();
15                 context.load(session);
16                 context.load(termStore);
17                 context.executeQueryAsync(
18                     function () {
19                         console.log('Got default term store');
20                     },
21                     function(sender, args) {
22                         console.log('Could not get default term store. ' + args.get_message());
23                     }
24                 );
25
26
27             });
28         });
29     });
30 });
31
32 var topnavbar;
33
34 topnavbar += '<div class="tbl-site-navigation">';
35 topnavbar += '<ul class="dropdown">';
36 topnavbar += '<li class=""><a href="#">The Brand Code - a</a></li>';
37 topnavbar += '<li class="dropdown1">';
38 topnavbar += '<ul class="sub_menu" style="visibility: hidden;">';
39 topnavbar += '<li class="large">';
40 topnavbar += '<div class="dropdownbox">';
41 topnavbar += '<div class="dropdownbox-title">Welcome to the Brand Code</div>';
42 topnavbar += '<ol>';
43 topnavbar += '<li><a href="">The importance of Brand Building</a></li>';
44 topnavbar += '<li><a href="">Introduction to the Brand Code</a></li>';
45 topnavbar += '<li><a href="">You and the Brand Code</a></li>';
46 topnavbar += '</ol>';
47 topnavbar += '</div>';

```

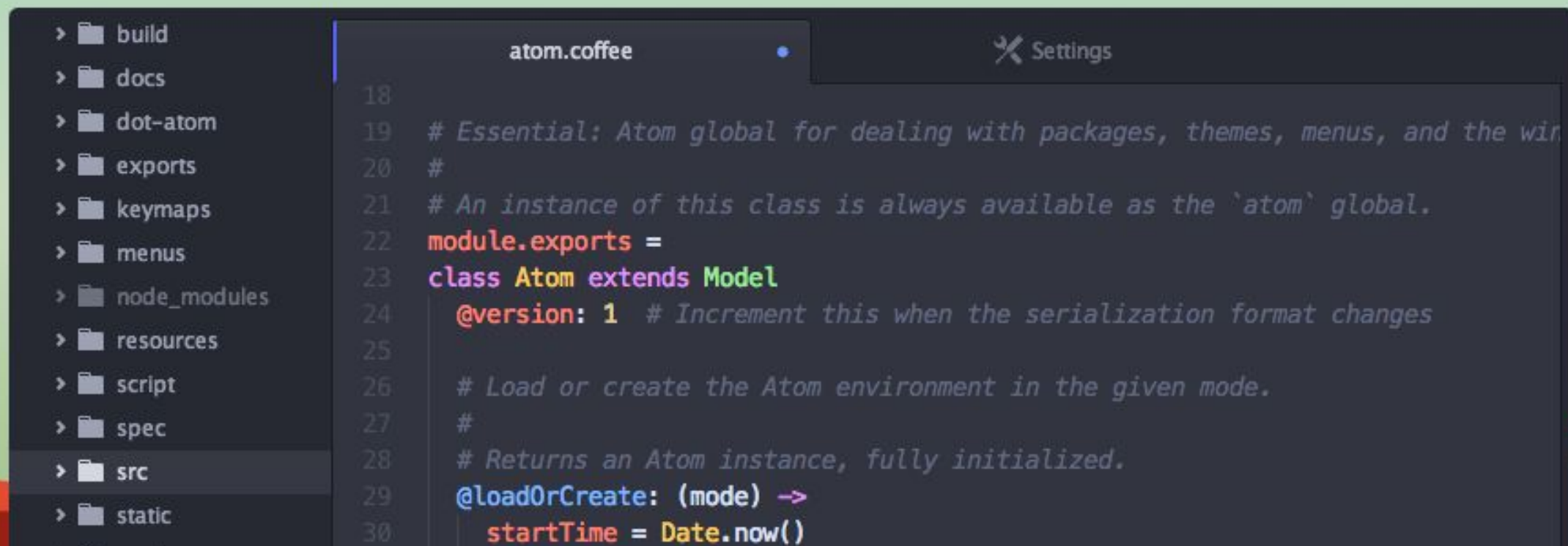
0 18

Ln 38, Col 72 UTF-8 CRLF JavaScript



ATOM

A hackable text editor
for the 21st Century



Reference

- Learn Git branching (interactive)
 - <http://pcottle.github.io/learnGitBranching/>
- Pro Git
 - <http://git-scm.com/book/>
- 寫給大家的 Git 教學
 - <http://www.slideshare.net/littlebtc/git-5528339>

Today's exercise

- Install Git command line tool in your computer.
 - Follow appendix “Git Command-line Tool Installation”.
- Try to submit in GitLab.